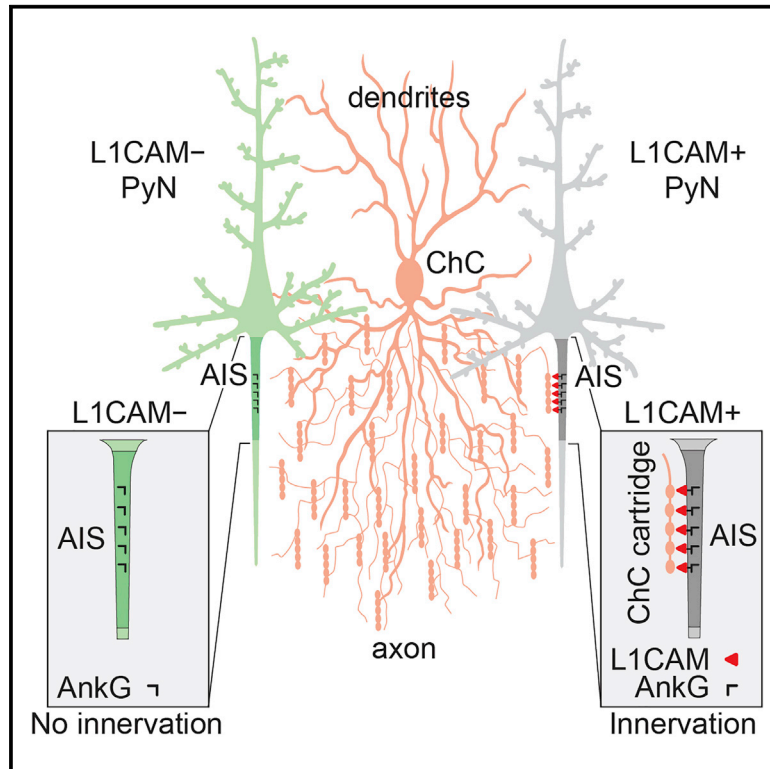


Neuron

Axo-axonic Innervation of Neocortical Pyramidal Neurons by GABAergic Chandelier Cells Requires AnkyrinG-Associated L1CAM

Graphical Abstract



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In Brief

Chandelier cells (ChCs) are the only interneuron subtype that selectively innervates the axon initial segment (AIS) of pyramidal neurons (PyNs) in the neocortex; yet, the underlying mechanisms are unknown. Tai et al. reveal that neocortical ChC/PyN AIS innervation requires ankyrin-G-clustered L1CAM.

Highlights

- NF186 is dispensable for neocortical PyN AIS innervation by ChCs
- Postsynaptic PyN-expressed L1CAM regulates PyN AIS synaptic innervation by ChCs
- L1CAM is required for the establishment and maintenance of ChC/PyN AIS innervation
- AnkG-mediated L1CAM anchoring at the AIS is necessary for ChC/PyN AIS innervation



Axo-axonic Innervation of Neocortical Pyramidal Neurons by GABAergic Chandelier Cells Requires AnkyrinG-Associated L1CAM

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SUMMARY

Among the diverse interneuron subtypes in the neocortex, chandelier cells (ChCs) are the only population that selectively innervate pyramidal neurons (PyNs) at their axon initial segment (AIS), the site of action potential initiation, allowing them to exert powerful control over PyN output. Yet, mechanisms underlying their subcellular innervation of PyN AISs are unknown. To identify molecular determinants of ChC/PyN AIS innervation, we performed an *in vivo* RNAi screen of PyN-expressed axonal cell adhesion molecules (CAMs) and select Ephs/ephrins. Strikingly, we found the L1 family member L1CAM to be the only molecule required for ChC/PyN AIS innervation. Further, we show that L1CAM is required during both the establishment and maintenance of innervation, and that selective innervation of PyN AISs by ChCs requires AIS anchoring of L1CAM by the cytoskeletal ankyrin-G/ β IV-spectrin complex. Thus, our findings identify PyN-expressed L1CAM as a critical CAM required for innervation of neocortical PyN AISs by ChCs.

INTRODUCTION

Proper assembly and functioning of cortical circuits rely on the formation of specific synaptic connections between excitatory pyramidal neurons (PyNs) and different types of GABAergic interneurons (Bartolini et al., 2013; Huang et al., 2007; Kepecs and Fishell, 2014). At least ten GABAergic interneuron subtypes have been identified in the cerebral cortex, each with uniquely organized axonal arbors that selectively innervate distinct subcellular compartments to control the input, integration, and output of their target cells (DeFelipe et al., 2013; Tremblay et al., 2016). Among them, chandelier cells (ChCs), also referred to as axo-axonic cells, are arguably the most distinctive (Howard

et al., 2005; Inan and Anderson, 2014; Jones, 1975; Somogyi, 1977; Szentágothai and Arbib, 1974; Woodruff et al., 2010). These cells, which predominantly derive from the ventral medial ganglionic eminence (vMGE) during late gestation (Inan et al., 2012; Taniguchi et al., 2013), exhibit a characteristic highly branched axon with multiple arrays of vertically oriented terminals, called cartridges, each harboring a string of synaptic boutons (Inan et al., 2007). Importantly, unlike other cortical interneurons that form somatodendritic synapses, ChC cartridges, typically 3–4 from 3–4 distinct ChCs, selectively innervate individual PyNs at their axon initial segment (AIS), the site of action potential initiation (DeFelipe et al., 1985; Somogyi, 1977). Furthermore, cartridges of single ChCs innervate hundreds of PyNs, which, combined with their exquisite subcellular specificity, makes them ideally suited to exert powerful control over PyN spiking and population output (DeFelipe et al., 1985; Howard et al., 2005; Inan et al., 2013; Woodruff et al., 2010). In line with this, recent studies have shown a critical role for ChCs in the synchronization of firing patterns of large populations of PyNs in different functional states (Glickfeld et al., 2009; Lu et al., 2017; Viney et al., 2013; Woodruff et al., 2011; Zhu et al., 2004). The importance of proper ChC function is further underscored by the association of ChC connectivity defects with brain disorders such as schizophrenia, epilepsy, and autism spectrum disorder (Ariza et al., 2018; Del Pino et al., 2013; Lewis, 2011; Ribak, 1985; Rocco et al., 2017). To date, however, the molecular mechanisms governing neocortical ChC/PyN AIS innervation remain entirely unknown. This has largely been due to the scarcity of ChCs and, most importantly, lack of unique ChC biochemical markers. Only recently have transgenic *Nkx2.1-Cre* mice that enable the reliable labeling of ChCs in the neocortex become available (Taniguchi et al., 2013; Xu et al., 2008).

Increasing evidence from other GABAergic interneuron subtypes indicates that the subcellular compartmentalization of synapses on principal neurons involves genetically determined mechanisms (Ango et al., 2004; Ashrafi et al., 2014; Di Cristo et al., 2004). In particular, cell adhesion molecules (CAMs) are emerging as key players in the axonal subcellular targeting of interneurons and the innervation of their postsynaptic cells (Ango

et al., 2004; Ashrafi et al., 2014; Guan and Maness, 2010; Telley et al., 2016). For example, in the cerebellum, the L1 immunoglobulin (Ig) CAM family member neurofascin-186 (NF186), which is present at the soma and AIS of Purkinje cells (PCs), directs the navigation of basket interneuron axons from the PC soma to the AIS, where it then facilitates pinceau synapse formation (Ango et al., 2004). In addition, recent work in the spinal cord found that an Ig CAM complex of NB2 and Caspr4 on sensory neurons interacts with the L1 Ig family members NrCAM and CHL1 on GABAergic interneurons to control synapse formation specifically at the axonal termini of sensory afferents (Ashrafi et al., 2014).

Given the above findings, we set out to perform *in vivo* screening of neocortical PyN-expressed axonal CAMs, including all those known to be enriched at the AIS, for their role(s) in ChC/PyN innervation (Ango et al., 2004; Hedstrom et al., 2007; Inda et al., 2006; Leterrier, 2016; Ogawa et al., 2008, 2010; Wimmer et al., 2015). In addition, as emerging evidence has implicated not only attractive but also repulsive mechanisms in the subcellular targeting/innervation of interneurons onto their target cells (Baohan et al., 2016; Brennaman et al., 2013; Kania and Klein, 2016), we tested select members of the Eph and ephrin family of receptors/ligands expressed in the neocortex. To do so, we devised a strategy taking advantage of *in utero* electroporation (IUE) and RNA interference (RNAi) to individually knock down these molecules in neocortical PyNs while concurrently labeling ChCs using the recently generated *Nkx2.1-CreER* mouse line (Taniguchi et al., 2013). Strikingly, of all the candidates tested, we found the pan-axonally expressed CAM L1CAM to be the only molecule required for neocortical ChC/PyN AIS innervation, as knockdown of PyN L1CAM, but none of the other screened candidates, significantly reduced PyN AIS innervation by ChCs. Furthermore, we demonstrate a requirement for L1CAM during both the establishment and maintenance of neocortical ChC/PyN AIS innervation. Finally, we provide evidence that anchoring of L1CAM at the AIS by the ankyrin-G/ β IV-spectrin AIS cytoskeletal complex is essential for ChC subcellular innervation of PyN AISs. Taken together, our findings identify L1CAM as the only PyN-expressed CAM known to date to regulate axo-axonic innervation of PyNs by ChCs in the neocortex.

RESULTS

AIS-Enriched CAM NF186 Is Dispensable for Neocortical PyN AIS Innervation by ChCs

Given the unique subcellular innervation of PyN AISs by ChCs, we reasoned that a CAM localized at the AIS of PyNs likely mediates this selective form of synaptic targeting in the neocortex. To explore this, we initially focused our attention on the AIS-enriched CAM NF186, given its previously demonstrated roles in governing GABAergic innervation of cerebellar PC AISs (Ango et al., 2004) and in clustering inhibitory postsynaptic proteins at the AIS of hippocampal granule cells (GCs) (Kriebel et al., 2011).

To determine whether NF186 is required for neocortical ChC/PyN AIS innervation, we combined IUE with RNAi in *Nkx2.1-CreER; Rosa26-loxPSTOPloxP-tdTomato* (*Ai9*) mice (Taniguchi et al., 2013) to knock down PyN-expressed NF186 and label

ChCs in the same neocortical layer (Figure 1A). In brief, vectors coexpressing EGFP and miR30-based short hairpin RNAs (shRNAs) capable of efficiently depleting NF186 protein levels (miR.NF#1 and #2) were generated (Figures S1A and S1B) and independently transfected into nascent layer II/III PyNs in *Nkx2.1-CreER;Ai9* animals via IUE at embryonic day 15.5 (E15.5). Then, tamoxifen (TMX) was administered to the embryos at E18.5 via oral gavage of the pregnant mother to trigger expression of tdTomato red fluorescent protein (RFP) in a sparse population of layer II ChCs (Figure 1A; Taniguchi et al., 2013). Brains from electroporated *Nkx2.1-CreER;Ai9* animals were collected at postnatal day 28 (P28), when ChCs are fully matured, sectioned coronally, and immunostained with an anti-ankyrin-G (AnkG) antibody to label the AIS. Then, confocal images were acquired, and the average percentage of GFP+ PyNs innervated at their AIS by single RFP+ ChCs in 200 μ m $\times$ 200 μ m fields of view (FOV) (50- μ m depth) in the somatosensory cortex was calculated (see STAR Methods and Figure S5 for details).

Surprisingly, we found that knockdown of PyN NF186 using either miR.NF#1 or miR.NF#2 had no effect on ChC/PyN AIS innervation (Figures 1B and 1C). In both groups, the percentage of GFP+ PyN AISs innervated by single RFP+ ChCs was similar to that observed in the control (miR.Ctrl) group (Figures 1B and 1C), implying that NF186 is not involved in neocortical ChC/PyN AIS innervation. To further corroborate these findings given NF186's demonstrated roles at the AIS of PCs and GCs in the cerebellum and hippocampus, respectively (Ango et al., 2004; Kriebel et al., 2011), we combined the CRISPR/Cas9 system (Cong et al., 2013) and IUE to selectively knockout NF186 in layer II/III neocortical PyNs. Specifically, we constructed a pX330 CRISPR plasmid expressing Cas9 and a single-guide RNA (sgRNA) against the *NF186* gene (crNF186) and electroporated this plasmid together with an EGFP-expressing vector into *Nkx2.1-CreER;Ai9* embryos at E15.5. Loss of NF186 protein in layer II/III GFP+ PyNs was verified by immunohistochemistry (Figure 1D), and the percentage of GFP+, NF186- PyN AISs innervated by single RFP+ ChCs was subsequently calculated. Consistent with the results obtained for PyN NF186 knockdown, CRISPR/Cas9-mediated knockout of NF186 in PyNs did not affect ChC/PyN AIS innervation (Figures 1D and 1E). Together, these data unequivocally demonstrate that NF186 is dispensable for proper neocortical PyN AIS innervation by ChCs, indicating that another/other molecule(s) must govern this unique form of axo-axonic innervation in the neocortex.

RNAi Screen of PyN-Expressed Cell Surface Molecules Identifies L1CAM as a Regulator of Neocortical ChC/PyN AIS Innervation

To identify the molecular factors required for neocortical ChC/PyN AIS innervation, we initiated an *in vivo* RNAi screen of PyN-expressed cell surface molecules present in the neocortex using shRNAs. We began by screening additional members of the L1 family of Ig CAMs (NrCAM, L1CAM, and CHL1) (Maness and Schachner, 2007) as well as other previously reported AIS-enriched adhesion molecules (TAG-1, SCN1B, ADAM22, and CASPR2) present in the neocortex (Figure 2A and Table S1)

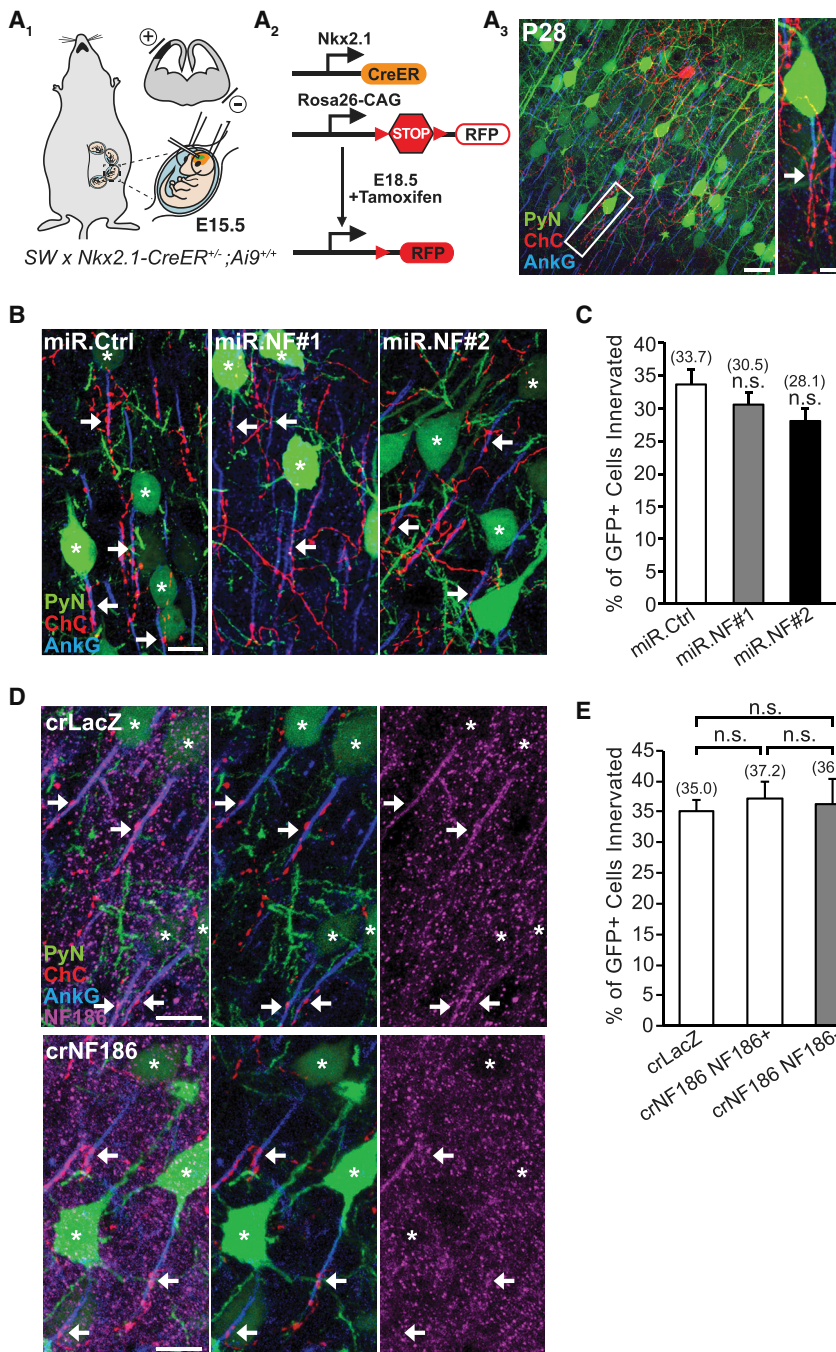


Figure 1. PyN AIS-Enriched NF186 Is Dispensable for Neocortical ChC/PyN AIS Innervation

(A) Experimental strategy to target/manipulate PyN gene expression and label ChCs in the same neocortical layer. (A₁) Schematic drawing of E15.5 *in utero* electroporation (IUE) targeting nascent layer II/III (LI/IIII) PyNs in embryos from Swiss Webster (SW) females that were bred with *Nkx2.1-CreER^{+/+}; Rosa26-loxpSTOPloxP-tdTomato (Ai9)^{+/+}* males is shown. The position of the positive (+) and negative (-) electrodes used to target neocortical progenitors in the ventricular zone (VZ) is depicted. (A₂) Tamoxifen (TMX) administration at E18.5 induces Cre activity and excision of a STOP cassette resulting in tdTomato red fluorescent protein (RFP) expression in ChC progenitors. (A₃) Representative 200 $\mu\text{m} \times 200 \mu\text{m}$ confocal image of a single RFP+ ChC and neighboring electroporated GFP+ PyNs in LII of somatosensory cortex is shown. Scale bar, 20 μm . Enlarged view of the boxed area showing a GFP+ PyN innervated at its AIS by an RFP+ ChC cartridge (arrow) is depicted on the right. Scale bar, 5 μm . AISs are visualized by immunostaining for ankyrin-G (AnkG) (blue).

(B) Representative images of PyNs innervated by ChC cartridges in LII of the somatosensory cortex from *Nkx2.1-CreER; Ai9* mice electroporated at E15.5 with plasmids expressing EGFP and miR30-based shRNAs against NF186 (miR.NF#1 or miR.NF#2) or *Renilla* luciferase (miR.Ctrl) and sacrificed at P28. Scale bar, 10 μm .

(C) Quantification of the percentage of GFP+ PyNs innervated by single RFP+ ChCs at P28. Innervation percentages are indicated for each condition in (C) and (E). 6–8 ChCs and 37–131 GFP+ PyNs per ChC from 3 animals were analyzed for each condition (one-way ANOVA, post hoc Tukey-Kramer test).

(D) Representative images of PyNs innervated by ChC cartridges in LII of somatosensory cortex from *Nkx2.1-CreER; Ai9* mice coelectroporated at E15.5 with plasmids expressing EGFP and Cas9 together with an sgRNA against *NF186* (crNF186) or *LacZ* (crLacZ), as a control, and sacrificed at P28. Scale bars, 10 μm . Endogenous NF186 is visualized by immunostaining for NF186 (magenta).

(E) Quantification of the percentage of GFP+ PyNs innervated by single RFP+ ChCs at P28. Of note, crNF186-electroporated PyNs still expressing NF186 (crNF186 NF186+) were also analyzed and included as an additional control. 8–9 ChCs and 17–73 GFP+ PyNs per ChC from 3 animals were analyzed for each condition (one-way ANOVA, post hoc Tukey-Kramer test).

For all images of ChC/PyN AIS innervation, stars and arrows indicate GFP+ PyNs innervated by RFP+ ChC cartridges and the site of GFP+ PyN AIS innervation, respectively. n.s. (not significant) indicates $p \geq 0.05$. Data are mean $\pm$ SEM. See also Figure S1.

(Inda et al., 2006; Ogawa et al., 2008; Ogawa et al., 2010; Wimmer et al., 2015). Furthermore, we also tested select Eph receptors and ephrins (Efn) (Figure 2A and Table S1) due to their well-documented roles in mediating attraction, repulsion, and/or adhesion through interactions with their respective ligand(s)/receptor(s) (EphA3, EphA4, EphB2, EfnB1, EfnB2, and EfnB3) (Kania and Klein, 2016).

For the screening of the above molecules, we used a strategy similar to the one described for NF186. Namely, vectors expressing EGFP and validated shRNAs targeting individual mouse cell surface molecules were constructed (Figures S1C–H and S1L–N; Anderson et al., 2012; Cissé et al., 2011; Khodosevich et al., 2011; McClelland et al., 2009; Nishikimi et al., 2011) and then individually transfected into nascent layer II/III PyNs of

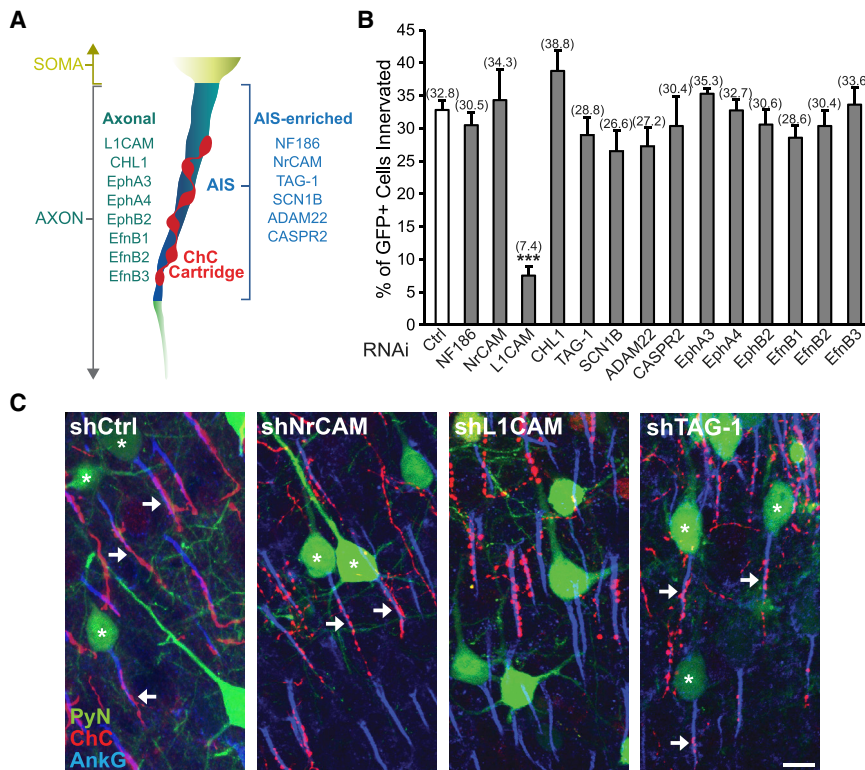


Figure 2. RNAi Screen of PyN-Expressed Cell Surface Molecules Identifies L1CAM as a Regulator of Neocortical ChC/PyN AIS Innervation

(A) Schematic depicting the subcellular localization (i.e., axonal or AIS enriched) of screened cell surface molecules in neocortical PyNs.

(B) In vivo RNAi screen of PyN-expressed axonal and AIS-enriched cell adhesion molecules and select members of the Eph and ephrin family of receptors/ligands present in the neocortex. Quantification of the percentage of GFP+ PyNs innervated by single RFP+ ChCs at P28 in LII of somatosensory cortex from *Nkx2.1-CreER;Ai9* mice electroporated at E15.5 with plasmids expressing EGFP and shRNAs targeting indicated cell surface molecules is shown. Innervation percentages are indicated for each condition. 6–16 ChCs and 12–99 GFP+ PyNs per ChC from 3 animals were analyzed for each condition (one-way ANOVA, post hoc Tukey-Kramer test).

(C) Representative images of PyNs innervated by ChC cartridges in LII of somatosensory cortex from *Nkx2.1-CreER;Ai9* mice electroporated at E15.5 with plasmids expressing EGFP and shCtrl, shNrCAM, shL1CAM, or shTAG-1 and sacrificed at P28. Scale bar, 10 μ m.

***p < 0.001. Data are mean $\pm$ SEM. See also Figure S1 and Table S1.

E15.5 *Nkx2.1-CreER;Ai9* embryos by IUE prior to TMX administration at E18.5. Electroporated brains were collected at P28, immunostained for AnkG, and imaged using a confocal microscope, and the average percentage of GFP+ PyNs innervated at their AIS by single RFP+ ChCs was calculated for each condition. Strikingly, of all the molecules tested, we found that only PyN knockdown of pan-axonally expressed L1CAM caused a marked reduction in PyN AIS innervation by ChCs (Figures 2B and 2C). Compared to the control shRNA group where approximately 33% of GFP+ PyNs were innervated by single RFP+ ChCs, only about 7% of GFP+ PyNs were innervated in the L1CAM shRNA group (Figure 2B). In all other groups, the percentage of GFP+ PyN AISs innervated by single RFP+ ChCs was not significantly different from that of the control group (Figures 2B and 2C). Together, these findings identify L1CAM as a strong candidate molecule critical for ChC/PyN AIS innervation in the neocortex.

Postsynaptic PyN-Expressed L1CAM Is Essential for Proper Neocortical PyN AIS Innervation by ChCs

Following our initial screening results, we set out to ensure that the observed impairment in ChC/PyN AIS innervation was due to specific knockdown of PyN L1CAM. To this end, we tested a second independent L1CAM shRNA (shL1CAM#2) as well as a miR30-based shRNA driven by the CAG promoter (CAG-miR.L1CAM). Both shRNAs substantially reduced endogenous L1CAM protein levels in cultured cortical neurons (Figures S1C and S1I). Consistent with the results obtained for the L1CAM shRNA used in our RNAi screen (shL1CAM#1), we found that expression of either shL1CAM#2 or CAG-miR.L1CAM in neocor-

tical PyNs significantly reduced the percentage of GFP+ PyN AISs innervated by single RFP+ ChCs (Figures 3A and 3B). In addition, we performed rescue experiments using a human L1CAM cDNA (hL1CAM) (Hlavin and Lemmon, 1991), which harbors three base pair mismatches with the mouse-targeted shL1CAM#1 sequence, making it resistant to shL1CAM#1-mediated L1CAM knockdown (Figure S1J). Coexpression of hL1CAM with shL1CAM#1 in neocortical PyNs completely rescued the RNAi-induced innervation phenotype (Figures 3A and 3B). Of note, expression of hL1CAM alone (hL1CAM overexpression [OE]) did not have any effect on ChC/PyN AIS synaptic innervation (Figures S2A–S2D). Combined, these data demonstrate that the effect of L1CAM RNAi on ChC/PyN AIS innervation is specific.

Next, we verified that the innervation phenotype caused by PyN L1CAM depletion was not an indirect consequence of alterations in PyN AIS structure and/or protein composition. To this end, we analyzed the length and area of the AIS as well as the distribution and levels of AIS-enriched proteins, including AnkG, β IV-spectrin, NF186, NrCAM, and Nav1.6, under PyN L1CAM knockdown conditions (Figures S3A–S3E). We found that none of these parameters were significantly different between the control and shL1CAM#1 groups, indicating that the L1CAM-dependent innervation phenotype did not result from altered PyN AIS structure/composition. Next, to rule out that the observed phenotype was secondary to defects in L1CAM-associated neurodevelopmental processes, such as neuronal migration and axon formation/guidance (Appel et al., 1993; Cohen et al., 1998; Lemmon et al., 1989; Lindner et al., 1983), we examined whether silencing of PyN L1CAM at a later

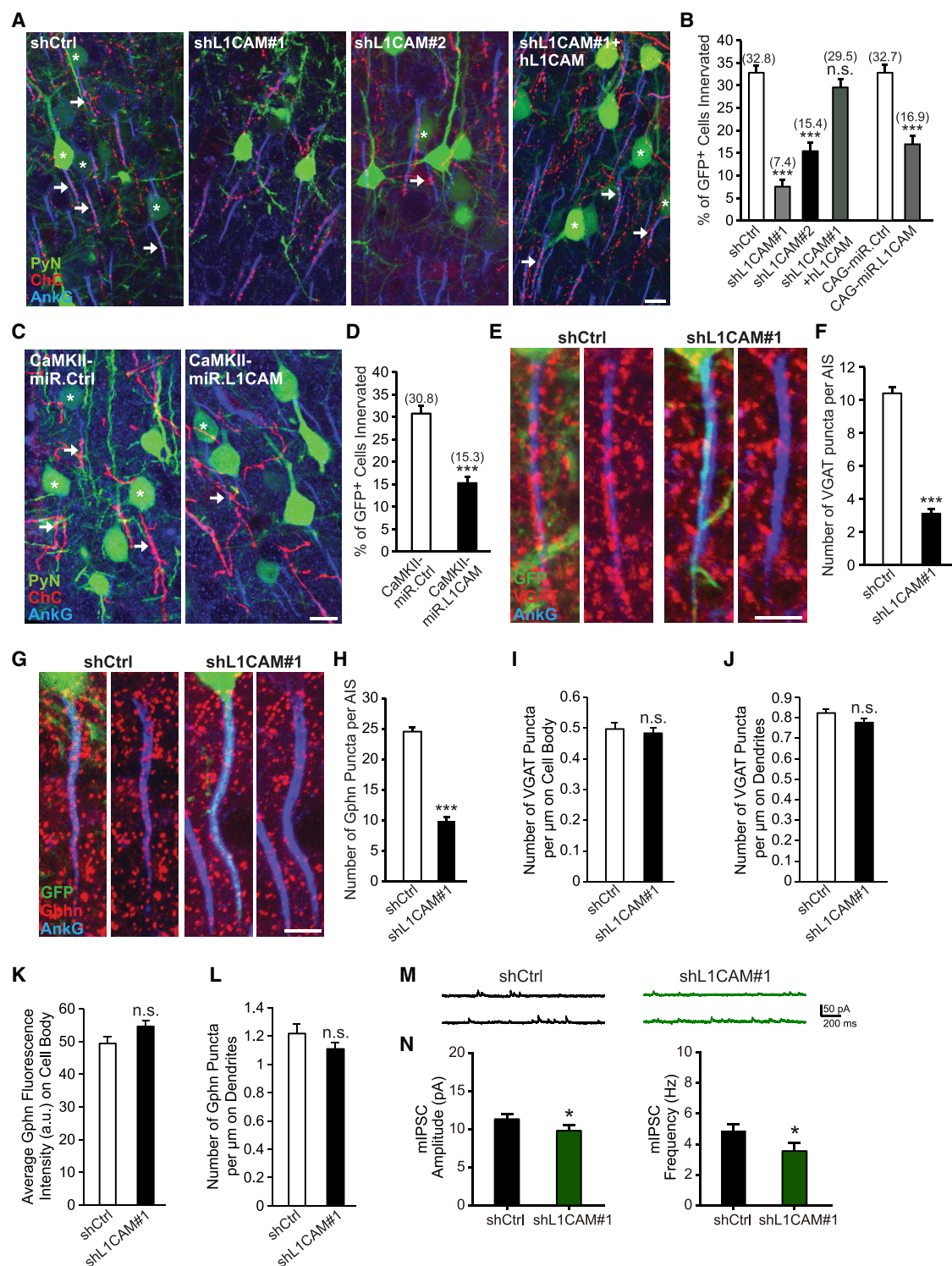


Figure 3. PyN L1CAM Is Required for Neocortical PyN AIS Synaptic Innervation by ChC Cartridges

(A) Representative images of PyNs innervated by ChC cartridges in LII of the somatosensory cortex from *Nkx2.1-CreER;Ai9* mice electroporated at E15.5 with plasmids expressing EGFP and shCtrl, shL1CAM#1, shL1CAM#2, or shL1CAM#1 + human L1CAM (hL1CAM) and sacrificed at P28. Scale bar, 10 μ m. (B) Quantification of the percentage of GFP+ PyNs innervated by single RFP+ ChCs at P28. Innervation percentages are indicated for each condition in (B) and (D). 9–16 ChCs and 17–126 GFP+ PyNs per ChC from 3 animals were analyzed for each condition (one-way ANOVA, post hoc Tukey-Kramer test [left] and Student's t test [right]).

(legend continued on next page)

developmental stage affects ChC/PyN AIS innervation. To accomplish this, we generated a vector that coexpresses EGFP and the miR30-based L1CAM shRNA from the CaMKII promoter (CaMKII-miR.L1CAM) (Figure S1K). This promoter becomes active in the second postnatal week, at which time the above neurodevelopmental processes are largely completed (data not shown). Analysis of electroporated brain slices at P28 revealed that CaMKII-driven L1CAM knockdown caused a significant reduction in ChC/PyN AIS percent innervation (Figures 3C and 3D), indicating that the L1CAM RNAi-induced innervation phenotype was not secondary to PyN L1CAM-dependent defects in earlier neurodevelopmental processes. Finally, to ensure that the observed innervation phenotype was not due to a reduction in ChC cartridge number, we counted the number of cartridges of single RFP+ ChCs in 200 $\mu\text{m} \times 200 \mu\text{m}$ FOVs (50- μm depth) in the somatosensory cortex of P28 *Nkx2.1-CreER;Ai9* animals electroporated at E15.5 with shCtrl- or shL1CAM#1-expressing constructs. No difference in ChC cartridge number was found between the shCtrl and shL1CAM#1 groups (Figure S3F). Together, these data unveil that PyN-expressed L1CAM is indispensable for neocortical PyN AIS innervation by ChCs.

PyN-Expressed L1CAM Selectively Regulates ChC/PyN GABAergic Synapses at the AIS

To corroborate and extend our above findings, we performed immunohistochemistry on brain slices with antibodies against AnkG and vesicular GABA transporter (VGAT), a presynaptic marker for GABAergic synaptic vesicles, or gephyrin, a postsynaptic marker for GABAergic synapses, to visualize axo-axonic synapses made by ChCs onto the AIS of PyNs. In particular, brain sections from P28 animals electroporated at E15.5 with vectors expressing EGFP and shCtrl or shL1CAM#1 were coimmunostained for AnkG and VGAT or gephyrin, and the number of VGAT- or gephyrin-positive puncta per GFP+ AIS was calculated. In line with our findings for ChC/PyN AIS innervation, we found that the number of VGAT and gephyrin puncta at the AIS of GFP+ PyNs was markedly reduced in the shL1CAM#1 group as compared to the shCtrl group (Figures 3E–3H). On the other

hand, the number of VGAT and gephyrin puncta along the somatodendritic compartment of GFP+ PyNs was not altered in the shL1CAM#1 group relative to the shCtrl group (Figures 3I–3L), implying that PyN L1CAM selectively regulates ChC/PyN AIS synapses, but not the synaptic innervation of other PyN subcellular domains by other interneuron subtypes. In this regard, it is also worth noting that knockdown of L1CAM in PyNs did not induce off-target ChC/PyN innervation at the somatodendritic compartment or distal axon of such shL1CAM#1-electroporated GFP+ PyNs (Figure S3G). Instead, under these knockdown conditions, neighboring non-electroporated GFP– PyNs with wild-type (WT) levels of L1CAM were preferentially innervated selectively at their AISs by ChCs (Figure S3H). In the same vein, we found that ectopic expression of hL1CAM in neocortical PyNs, which failed to induce any changes in ChC/PyN AIS percent innervation, also did not drive ChC/PyN innervation outside the AIS (Figure S2B).

Given our findings demonstrating a marked reduction in GABAergic synapse number at the AIS of L1CAM-depleted PyNs in addition to recent reports demonstrating ChC-mediated inhibition of PyN firing (Lu et al., 2017; Woodruff et al., 2011), we investigated whether inhibitory synaptic transmission onto L1CAM-depleted PyNs is altered. To this end, whole-cell patch-clamp recordings were performed to measure miniature inhibitory postsynaptic currents (mIPSCs) in layer II/III GFP+ PyNs in the somatosensory cortex of P27–P30 mice electroporated at E15.5 with plasmids expressing EGFP and shCtrl or shL1CAM#1. Interestingly, we found that both the frequency and amplitude of mIPSCs were decreased in the shL1CAM#1 group relative to the shCtrl group (Figures 3M and 3N). Combined with our findings that the number of GABAergic synapses on PyN AISs, but not along their somatodendritic compartment, is decreased under PyN L1CAM knockdown conditions, these results suggest that silencing of PyN L1CAM reduces the inhibitory input from ChCs onto L1CAM-depleted PyNs.

Altogether, these data establish PyN-expressed L1CAM as a critical factor required for proper neocortical ChC/PyN AIS GABAergic synaptic innervation. To the best of our knowledge,

(C) Representative images of PyNs innervated by ChC cartridges in LII of somatosensory cortex from *Nkx2.1-CreER;Ai9* mice electroporated at E15.5 with plasmids expressing EGFP and CaMKII-miR.Ctrl or CaMKII-miR.L1CAM and sacrificed at P28. Scale bar, 10 μm .

(D) Quantification of the percentage of GFP+ PyNs innervated by single RFP+ ChCs at P28. 10–14 ChCs and 32–147 GFP+ PyNs per ChC from 3 animals were analyzed for each condition (Student's *t* test).

(E and G) Representative images of GFP+ PyN AISs in LII of somatosensory cortex from CD1 mice electroporated at E15.5 with plasmids expressing EGFP and shCtrl or shL1CAM#1 and sacrificed at P28. Inhibitory synapses are visualized by immunostaining for the GABAergic presynaptic marker VGAT (red; E) or the GABAergic postsynaptic marker gephyrin (Gphn; red; G). Scale bars, 5 μm .

(F and H) Quantification of the average number of VGAT (F) or gephyrin (H) puncta per GFP+ PyN AIS at P28. 33–43 AISs from 3 animals were analyzed for each condition (Student's *t* tests).

(I) Quantification of the average number of VGAT puncta per μm on GFP+ PyN cell bodies at P28. 30 cell bodies from 3 animals were analyzed for each condition (Student's *t* test).

(J and L) Quantification of the average number of VGAT (J) or gephyrin (L) puncta per μm on the dendrites of GFP+ PyNs at P28. 24–51 dendrites from 3 animals were analyzed for each condition (Student's *t* tests).

(K) Quantification of the average fluorescence intensity (a.u.) of gephyrin on the cell body of GFP+ PyNs at P28. 34 cell bodies from 3 animals were analyzed for each condition (Student's *t* test).

(M) Representative traces of mIPSCs recorded from GFP+ PyNs in LII/III of somatosensory cortex from CD1 mice electroporated at E15.5 with plasmids expressing EGFP and shCtrl or shL1CAM#1 and sacrificed at P27–P30.

(N) Quantification of mIPSC amplitude (left) and frequency (right). 16 GFP+ PyNs from 4 animals were analyzed for each condition (Student's *t* tests).

n.s. indicates $p \geq 0.05$, * $p < 0.05$, *** $p < 0.001$. Data are mean $\pm$ SEM. See also Figures S1–S3.

L1CAM is so far the only PyN-expressed adhesion molecule identified to date to govern PyN AIS innervation by ChCs in the neocortex.

L1CAM Is Required for Both the Establishment and Maintenance of Neocortical ChC/PyN AIS Innervation

To further define the role of L1CAM in neocortical ChC/PyN AIS innervation, we asked when L1CAM is required during the innervation process. Namely, is it required early on for the establishment and/or later on for the maintenance of this unique form of subcellular innervation? First, to address this, we needed to determine the temporal profile of neocortical PyN AIS innervation by ChCs. To this end, brains from E18.5 TMX-induced *Nkx2.1-CreER;Ai9* animals were collected at time points spanning P8 to P28, sectioned and coimmunostained for AnkG and gephyrin, and analyzed for the percentage of PyN AISs innervated by single RFP+ ChCs in 200 $\mu\text{m} \times 200 \mu\text{m}$ FOVs (Figures 4A and 4B). In addition, the number of gephyrin puncta per PyN AIS and the percentage of RFP+ ChC boutons overlapping with PyN AIS gephyrin puncta were quantified (Figures 4C and 4D).

Interestingly, we found that there was minimal innervation of PyN AISs by ChCs in layer II/III of the somatosensory cortex from P8 to P11 but that a sharp increase in innervation occurred at P12 (Figures 4A and 4B). Notably, this sharp increase was accompanied by a simultaneous rise in the average number of gephyrin puncta per PyN AIS and the percentage of AIS-associated RFP+ ChC boutons overlapping with gephyrin puncta (Figures 4A, 4C, and 4D). In line with this, we found that such a spike in ChC/PyN AIS innervation paralleled the development of neocortical ChC axons, which undergo a marked increase in branching/complexity starting at P11/P12 (Figures 4E–4I). The increase in ChC/PyN AIS innervation and gephyrin puncta per PyN AIS continued into the third/fourth postnatal week before stabilizing at approximately P28 (Figures 4B and 4C). Importantly, the increase in the average number of gephyrin puncta per PyN AIS starting at P12 likely reflects the establishment of true ChC/PyN synapses, as the percentage of AIS-associated RFP+ ChC boutons overlapping with gephyrin puncta approached 100% by P13/P14 (Figure 4D). Together, these data reveal that neocortical ChC/PyN AIS innervation starts at approximately P12 and reaches final maturity by P28.

Next, we examined whether L1CAM expression in PyNs is correlated with and required early on during the onset of ChC/PyN AIS innervation. To this end, we began by performing coimmunostainings of L1CAM and AnkG on mouse brain cryosections at time points spanning P10 to P13. Significantly, we found L1CAM expression to increase at the AIS of layer II/III neocortical PyNs from P10 to P13 (Figures 5A and 5B), coinciding with the time window, within which ChCs first start to innervate the AIS of PyNs. Moreover, as a complementary approach, we took advantage of the novel genome-editing technique SLEND (single-cell labeling of endogenous proteins by CRISPR/Cas9-mediated homology-directed repair) (Mikuni et al., 2016) to insert an hemagglutinin-encoding (HA-encoding) epitope tag into the genomic locus of *L1CAM*. Single-cell genome editing in neocortical PyNs is achieved by delivering the editing machinery to dividing PyN progenitors via IUE. By coimmunostaining for HA and AnkG in electroporated brain slices collected from P10

and P13 mice, we found a similar increase in L1CAM-HA expression at the AIS of transfected PyNs during this developmental time window (Figures 5C and 5D), further corroborating our *in vivo* L1CAM immunohistochemistry findings. Then, we investigated whether knockdown of PyN L1CAM affects AIS innervation at P14 when synaptic connections between ChCs and PyN AISs are just starting to form. Brains of either *Nkx2.1-CreER;Ai9* or CD1 mice electroporated at E15.5 with shL1CAM#1- or shCtrl-expressing vectors were collected at P14, sectioned and coimmunostained for AnkG with or without gephyrin, and analyzed as described above. Similar to our findings at P28, we observed a significant decrease in ChC/PyN AIS percent innervation and, correspondingly, in the average number of gephyrin puncta per GFP+ PyN AIS in the shL1CAM#1 group relative to the control group at P14 (Figures 5E–5H), suggesting that PyN L1CAM is needed during early postnatal development for the establishment of neocortical ChC/PyN AIS innervation.

Finally, we sought to determine whether L1CAM is also required for the maintenance of ChC/PyN AIS innervation at later postnatal stages. To do so, we engineered a Cre-dependent inducible RNAi expression vector targeted against L1CAM (FLIP-miR.L1CAM) that allows for the temporal control of L1CAM knockdown postnatally via TMX administration (Figures 5I–5J) (Tai et al., 2014). Of note, due to the inducible nature of the FLIP-miR.L1CAM construct, this experiment could not be performed in *Nkx2.1-CreER;Ai9* mice since TMX induction at E18.5 to label ChCs would turn on expression of miR.L1CAM embryonically. Hence, instead we employed a double IUE approach, separately targeting neocortical ChCs and PyNs in WT CD1 animals. Namely, in the same embryos, we performed vMGE-directed IUEs at E13.5 to label nascent layer II ChCs with a tdTomato fluorescent marker (Tai et al., 2014) and E15.5 ventricular-zone-directed (VZ-directed) IUEs to co-transfect nascent layer II/III PyNs with FLIP-miR.L1CAM, or FLIP-miR.Ctrl, and a vector expressing a TMX-inducible form of Cre recombinase ($\text{ER}^{\text{T2}}\text{CreER}^{\text{T2}}$; Figure 5K). Then, such doubly electroporated animals were given TMX at P28 to turn on expression of miR.L1CAM and their brains were collected at P40 for analysis. Interestingly, we found that both the percentage of GFP+ PyN AISs innervated by single RFP+ ChCs and the average number of gephyrin puncta per GFP+ PyN AIS were significantly reduced in the FLIP-miR.L1CAM group relative to the FLIP-miR.Ctrl group (Figures 5L–5O). Together, these data indicate that L1CAM is required not only during the establishment of neocortical ChC/PyN AIS innervation but also for the maintenance of such innervation into adulthood.

L1CAM Anchoring at the AIS via the Ankyrin-G/ β IV-Spectrin Cytoskeletal Complex Is Necessary for PyN AIS Innervation by ChCs

While our above findings demonstrate a critical role for PyN L1CAM in governing neocortical ChC/PyN AIS innervation, an outstanding question is how L1CAM mediates this unique form of subcellular innervation, given that it is expressed throughout the axon (Boiko et al., 2007; Scotland et al., 1998; Winckler et al., 1999). Due to the stark difference in ChC/PyN innervation between PyN axonal subdomains (AIS and distal axon), we postulated that AIS-localized L1CAM differs from the pool of

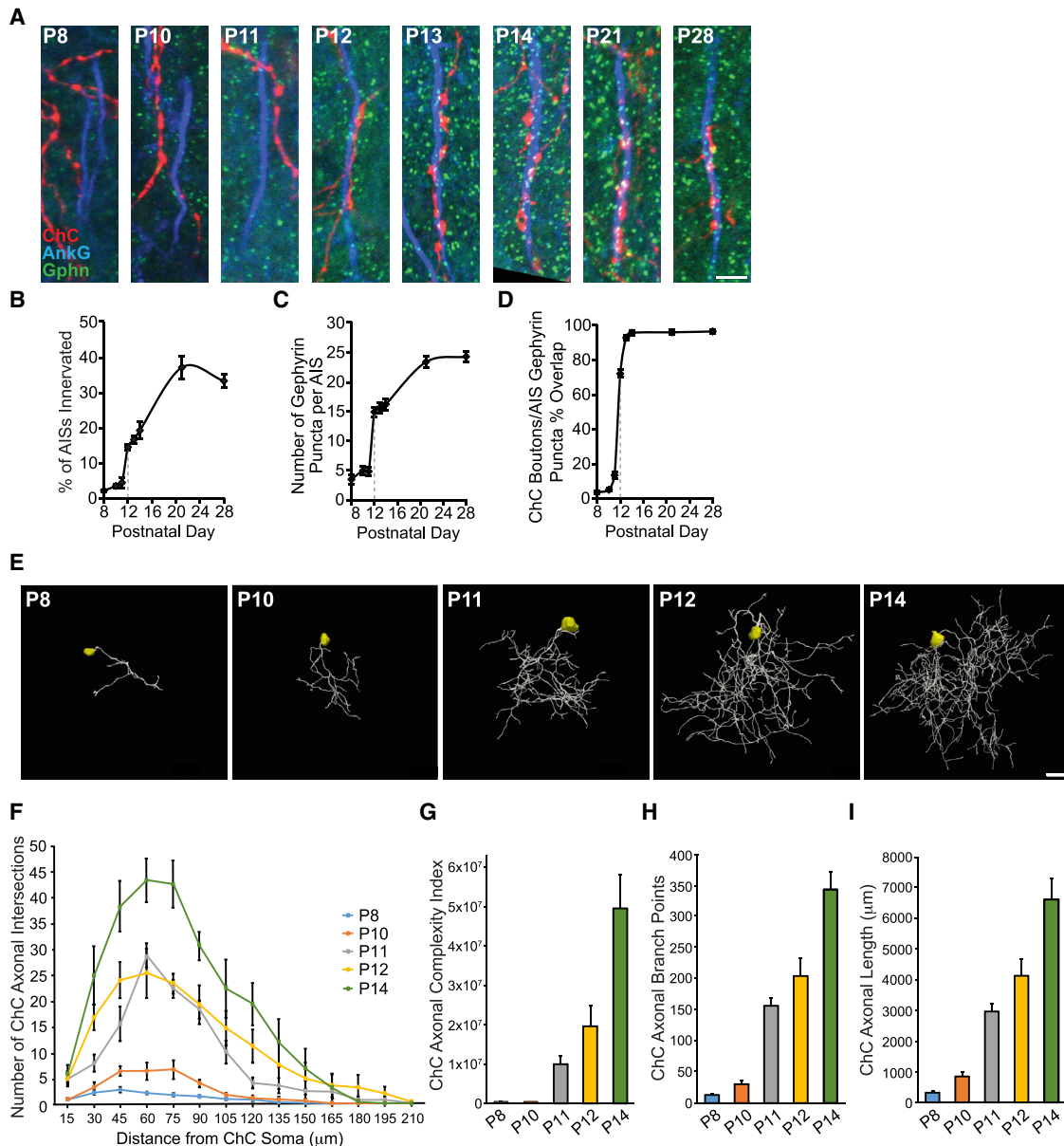


Figure 4. Developmental Profiling of ChC Axonal Morphology and Connectivity

(A) Temporal profile of PyN AIS innervation by ChCs in LII of somatosensory cortex from *Nkx2.1-CreER;Ai9* mice. Representative images of RFP+ ChC cartridges, AISs of neighboring PyNs, and gephyrin puncta at time points spanning P8 to P28 are shown. Scale bar, 5 μm .

(B–D) Quantification of the percentage of PyN AISs innervated by single RFP+ ChCs (B), the average number of gephyrin puncta per PyN AIS (C), and the percentage of RFP+ ChC boutons overlapping with AIS gephyrin puncta (D) from P8 to P28. 5–9 ChCs and 177–321 PyNs (B) or 8–25 PyN AISs (C and D) per ChC from 3 animals were analyzed for each time point.

(E) Representative maximum projection renderings from 3D reconstructions of individual neocortical RFP+ ChC somas and axons at time points spanning P8 to P14. Scale bar, 20 μm .

(F) Scholl analysis comparing axonal arbors of ChCs at time points spanning P8 to P14. 4–7 ChCs from 3 animals were analyzed for each time point.

(G–I) Morphometric analyses of ChC axonal complexity index (G), branch points (H), and length (I) at time points spanning P8 to P14. 4–7 ChCs from 3 animals were analyzed for each time point.

Data are mean $\pm$ SEM.

L1CAM in the distal axon via its binding to an AIS-specific protein(s). Of particular relevance in this regard is L1CAM's interaction with the AIS-localized scaffolding protein AnkG (Boiko et al.,

2007; Davis and Bennett, 1994; Hortsch et al., 2009; Needham et al., 2001), the master organizer of the AIS, which plays a critical role in linking AIS transmembrane components to the actin

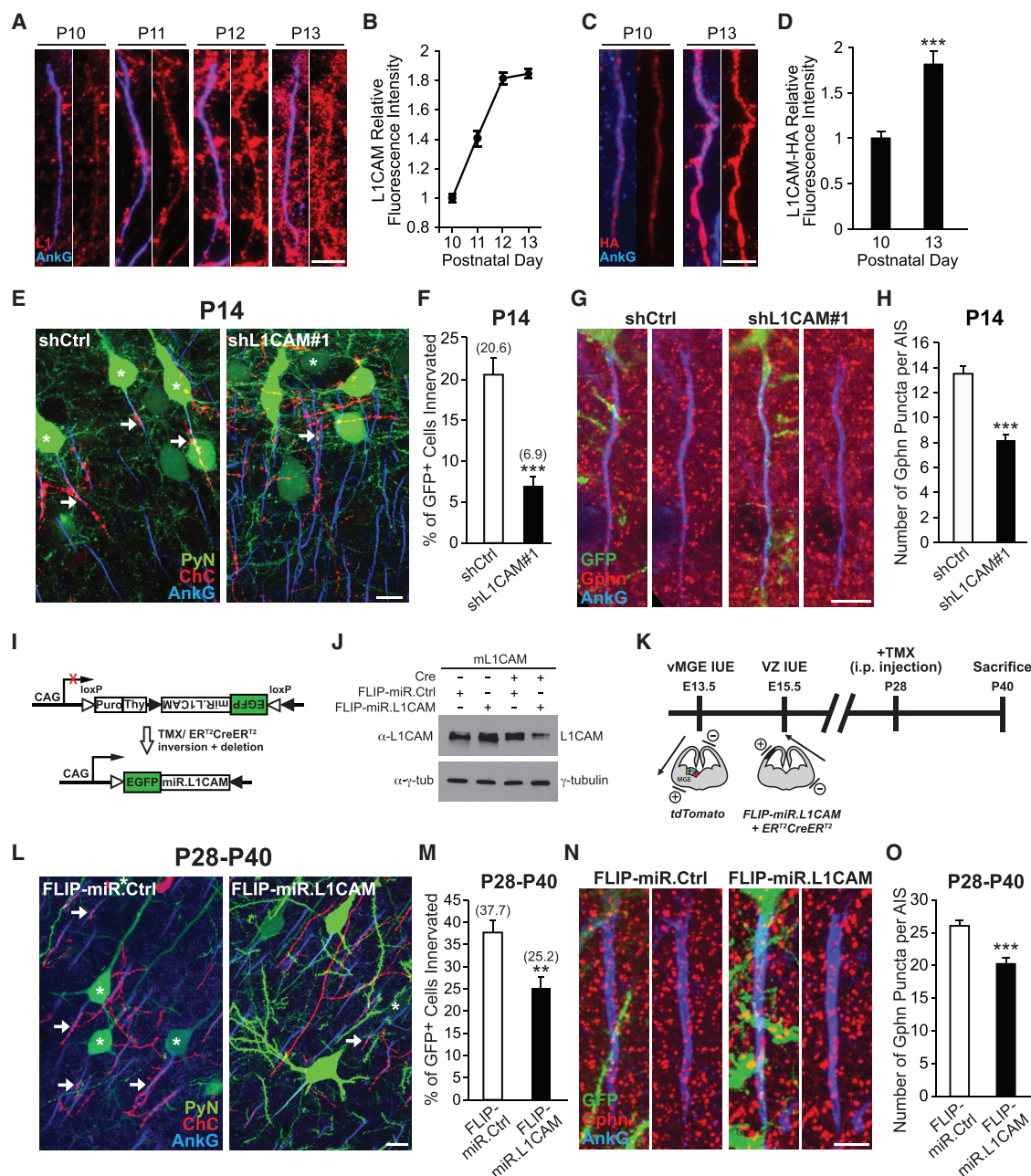


Figure 5. PyN L1CAM Is Required for the Establishment and Maintenance of Neocortical ChC/PyN AIS Innervation

(A) Temporal expression profile of endogenous L1CAM at the AIS of PyNs in LII of somatosensory cortex from CD1 mice. Representative images of PyN AISs at time points spanning P10–P13 are shown. L1CAM is visualized by immunostaining for L1CAM (L1; red). Scale bar, 5 μ m.

(B) Quantification of the relative fluorescence intensity of endogenous L1CAM at the AIS of PyNs at time points spanning P10–P13. 90–155 AISs from 3 animals were analyzed for each time point.

(C) Temporal expression profile of C-terminal HA-tagged endogenous L1CAM (L1CAM-HA) at the AIS of transfected PyNs in LII of somatosensory cortex from CD1 mice coelectroporated at E13.5 with pCAG-YFP, SLENDR construct pX330-crL1CAM, and ssODN-L1CAM. Representative images of electroporated PyN AISs at P10 and P13 are shown. L1CAM-HA is visualized by immunostaining for HA (red). Scale bar, 5 μ m.

(D) Quantification of the relative fluorescence intensity of L1CAM-HA at the AIS of electroporated PyNs at P10 and P13. 38–42 AISs from 3 animals were analyzed for each time point (Student's t test).

(E) Representative images of PyNs innervated by ChC cartridges in LII of somatosensory cortex from *Nkx2.1-CreER;Ai9* mice electroporated at E15.5 with plasmids expressing EGFP and shCtrl or shL1CAM#1 and sacrificed at P14. Scale bar, 10 μ m.

(F) Quantification of the percentage of GFP+ PyNs innervated by single RFP+ ChCs at P14. Innervation percentages are indicated for each condition in (F) and (M). 12–13 ChCs and 12–86 GFP+ PyNs per ChC from 3 animals were analyzed for each condition (Student's t test).

(legend continued on next page)

cytoskeleton via its interaction with β IV-spectrin (Leterrier, 2016; Yang et al., 2007). Moreover, previous work found AnkG to promote L1CAM oligomerization and limit its diffusibility selectively at the AIS *in vitro* (Boiko et al., 2007; Davis and Bennett, 1994; Winckler et al., 1999). Based on these findings and our data showing an enrichment of surface L1CAM at the AIS relative to the distal axon of primary cultured cortical PyNs (Figures S4A and S4E), we postulated that L1CAM-AnkG- β IV-spectrin interactions facilitate the clustering and enrichment of surface L1CAM at the AIS of PyNs, thereby enabling proper subcellular innervation by ChCs.

To test this possibility, we investigated whether perturbing L1CAM-AnkG- β IV-spectrin interactions affects the subcellular distribution of surface L1CAM and, most importantly, ChC/PyN AIS innervation. To this end, we utilized a previously described human L1CAM mutant containing a disease-linked mutation in its cytoplasmic FIGQY motif (Y1229H) that disrupts its interaction with AnkG (Needham et al., 2001). In addition, we used miR30-based shRNAs (miR. β IV#1 and #2) capable of efficiently reducing β IV-spectrin protein expression (Figure S4D) as another means to disrupt L1CAM-AnkG- β IV-spectrin interactions at the AIS. To assess the effects of the Y1229H mutation and β IV-spectrin knockdown on the subcellular distribution of surface L1CAM, we performed live cell surface labeling of L1CAM on primary cultured cortical PyNs. In line with a critical role for AnkG and β IV-spectrin in regulating the anchoring/clustering of L1CAM at the AIS, we found that the levels of surface L1CAM at the AIS were significantly reduced in neurons expressing L1CAM-Y1229H or β IV-spectrin shRNA, as compared to control neurons expressing L1CAM-WT or miR.Ctrl shRNA, respectively (Figures S4A–S4C and S4E–S4G).

Next, we tested the ability of the Y1229H mutant to rescue the L1CAM RNAi-induced defect in ChC/PyN AIS innervation. To this end, a vector expressing RNAi-resistant human (h)

L1CAM-Y1229H was co-transfected with shL1CAM#1 into nascent layer II/III PyNs of *Nkx2.1-CreER;Ai9* animals via E15.5 IUE, and the percentage of GFP+ PyN AISs innervated by single RFP+ ChCs was calculated at P14 and P28. We found that coexpression of hL1CAM-Y1229H with shL1CAM#1 failed to rescue the L1CAM RNAi-induced innervation defect at both P14 and P28, in contrast to control hL1CAM-WT (Figures 6A–6C). Finally, we examined the effects of PyN β IV-spectrin knockdown on ChC/PyN AIS innervation using the aforementioned miR. β IV#1 and #2 shRNAs. Expression of either miR. β IV#1 or miR. β IV#2 in layer II/III PyNs significantly reduced the percentage of GFP+ PyNs innervated by single RFP+ ChCs at P14 and P28 (Figures 6D–6F). Together, these data demonstrate that L1CAM's interaction with the AnkG- β IV-spectrin AIS cytoskeletal complex is critical for its role in mediating selective innervation of neocortical PyN AISs by ChCs.

DISCUSSION

The striking selectivity of ChC innervation at the AIS of PyNs has intrigued neuroscientists ever since the initial discovery of ChCs in the 1970s (Jones, 1975; Somogyi, 1977; Szentágothai and Arbib, 1974). This interest has heightened with accruing evidence showing that this unique form of subcellular innervation allows ChCs to exert powerful yet precise control over cortical PyN spiking and population output (Howard et al., 2005; Inan and Anderson, 2014; Lu et al., 2017) and, in particular, given increasing reports linking ChC connectivity defects to neurological conditions, such as schizophrenia and epilepsy (Ariza et al., 2018; Del Pino et al., 2013; Lewis, 2011; Ribak, 1985; Rocco et al., 2017). Despite these findings, little to nothing is known about the mechanisms underlying the subcellular innervation of neocortical PyN AISs by ChCs. While the atypical Rac activator DOCK7 and the receptor tyrosine kinase ErbB4 have been

(G) Representative images of GFP+ PyN AISs in LII of somatosensory cortex from CD1 mice electroporated at E15.5 with plasmids expressing EGFP and shCtrl or shL1CAM#1 and sacrificed at P14. Inhibitory synapses are visualized by immunostaining for gephyrin (red). Scale bar, 5 μ m.

(H) Quantification of the average number of gephyrin puncta per GFP+ PyN AIS at P14. 33–49 AISs from 3 animals were analyzed for each condition (Student's t test).

(I) Cre-dependent RNAi expression vector used for the temporal regulation of L1CAM knockdown, based on previously described "FLIP vectors" (Stern et al., 2008). Cre expression inverts the EGFP-miR.L1CAM sequence, allowing for its regulated expression, and at the same time deletes the Puro-2A-Thy1.1 encoding sequences.

(J) Knockdown of L1CAM using inducible L1CAM RNAi. Western blot of total lysates from HEK293T cells co-transfected with plasmids expressing mouse L1CAM (mL1CAM) and CAG-driven Cre, FLIP-miR.Ctrl, and/or FLIP-miR.L1CAM, probed with antibodies to L1CAM and γ -tubulin is shown. Data shown are representative of three independent experiments.

(K) Scheme of experimental workflow to test whether L1CAM is required for the maintenance of neocortical ChC/PyN AIS innervation. First, ChC progenitors in CD1 embryos were labeled via vMGE-directed IUE at E13.5 with a tdTomato-expressing plasmid. Then, the same embryos were subjected to VZ-directed IUE at E15.5 to introduce FLIP-miR.L1CAM- or FLIP-miR.Ctrl-expressing vectors together with a CAG-ER^{T2}CreER^{T2}-expressing plasmid into PyN progenitors. Positions of the positive (+) and negative (–) electrodes used during vMGE- and VZ-directed IUEs are depicted. Animals were administered TMX via intraperitoneal (i.p.) injection at P28 to induce L1CAM RNAi expression and sacrificed at P40.

(L) Representative images of PyNs innervated by ChC cartridges in LII of somatosensory cortex from CD1 mice subjected to dual IUEs, induced at P28 with TMX, and sacrificed at P40 as described in (K). Scale bar, 10 μ m.

(M) Quantification of the percentage of GFP+ PyNs innervated by single RFP+ ChCs at P40. 15–18 ChCs and 4–40 GFP+ PyNs per ChC from 3 animals were analyzed for each condition (Student's t test).

(N) Representative images of GFP+ PyN AISs in LII of somatosensory cortex from CD1 mice coelectroporated with pCAG-ER^{T2}CreER^{T2} and plasmids expressing FLIP-miR.L1CAM or FLIP-miR.Ctrl, TMX induced at P28, and sacrificed at P40. Inhibitory synapses are visualized by immunostaining for gephyrin (red). Scale bar, 5 μ m.

(O) Quantification of the average number of gephyrin puncta per GFP+ PyN AIS at P40. 27–30 AISs from 3 animals were analyzed for each condition (Student's t test).

p < 0.01, *p < 0.001. Data are mean $\pm$ SEM.

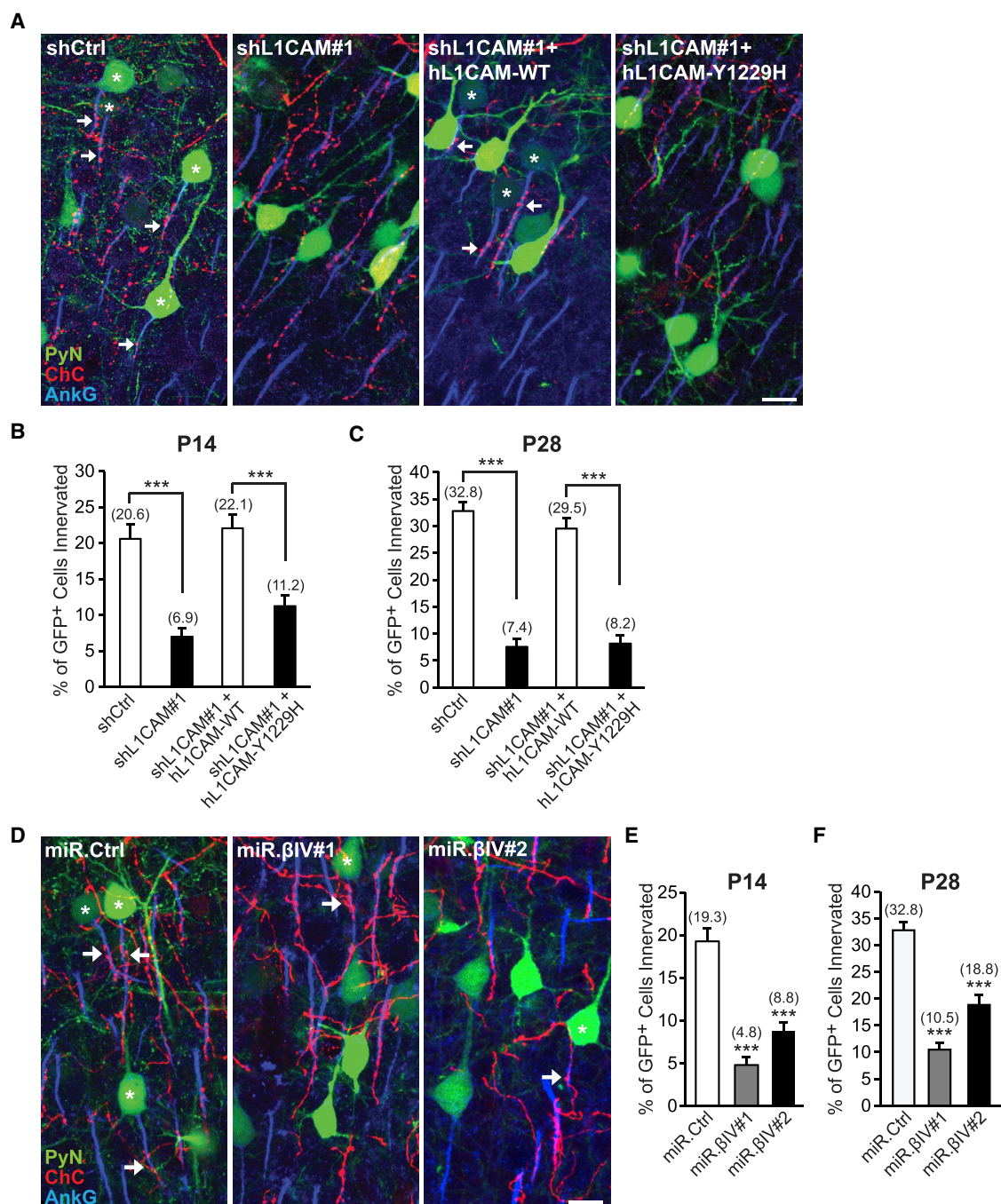


Figure 6. L1CAM's Interaction with the AnkG-βIV-Spectrin AIS Cytoskeletal Complex Is Necessary for PyN AIS Innervation by ChCs

(A) Representative images of PyNs innervated by ChC cartridges in LII of somatosensory cortex from *Nkx2.1-CreER;Ai9* mice electroporated at E15.5 with plasmids expressing EGFP and shCtrl, shL1CAM#1, shL1CAM#1+hL1CAM-WT, or shL1CAM#1+hL1CAM-Y1229H and sacrificed at P28. Scale bar, 10 μm.

(B and C) Quantification of the percentage of GFP+ PyNs innervated by single RFP+ ChCs at P14 (B) and P28 (C). Innervation percentages are indicated for each condition at each time point in (B), (C), (E), and (F). 10–16 ChCs and 12–110 GFP+ PyNs per ChC from 3 animals were analyzed for each condition (one-way ANOVAs, post hoc Tukey-Kramer tests).

(D) Representative images of PyNs innervated by ChC cartridges in LII of somatosensory cortex from *Nkx2.1-CreER;Ai9* mice electroporated at E15.5 with plasmids expressing EGFP and miR.Ctrl, miR.βIV#1, or miR.βIV#2 and sacrificed at P28. Scale bar, 10 μm.

(E and F) Quantification of the percentage of GFP+ PyNs innervated by single RFP+ ChCs at P14 (E) and P28 (F). 6–10 ChCs and 24–132 GFP+ PyNs per ChC from 3 animals were analyzed for each condition (one-way ANOVAs, post hoc Tukey-Kramer tests).

****p* < 0.001. Data are mean ± SEM. See also Figure S4.

implicated in the regulation of ChC cartridge bouton size/density (Fazzari et al., 2010; Tai et al., 2014), depletion of these proteins in ChCs does not affect the percentage of PyN AISs innervated by ChCs (Figures S6A–S6D). To identify the molecular factors required for ChC/PyN AIS innervation in the neocortex, we devised a strategy in this study to target/manipulate PyN gene expression and label ChCs in the same neocortical layer. This strategy enabled us to screen PyN-expressed axonal CAMs in addition to select Ephs and ephrins present in the neocortex for their involvement in ChC/PyN AIS innervation *in vivo*. Strikingly, our results revealed that silencing of only L1CAM, and none of the other molecules tested, caused a marked reduction in neocortical PyN AIS innervation by ChCs. In line with this, we found the number of VGAT and gephyrin puncta at the AIS, but not along the somatodendritic compartment, to be concomitantly reduced in L1CAM-depleted PyNs, indicating that PyN L1CAM selectively regulates ChC/PyN AIS synaptic innervation but not the subcellular targeting of other PyN subcellular domains by other interneuron subtypes. Our data further reveal that PyN L1CAM plays a dual role in this process, as it is required for both the establishment and maintenance of neocortical ChC/PyN AIS innervation postnatally. Regarding establishment, we found that (1) L1CAM expression at the AIS of PyNs markedly increases during the time window of initial ChC/PyN AIS innervation and (2) depleting PyN L1CAM perturbs ChC/PyN AIS innervation during the same time frame. Regarding maintenance, silencing of PyN L1CAM after the majority of ChC/PyN axo-axonic synapses have already been established still causes a significant decrease in ChC/PyN innervation and gephyrin puncta number per AIS. Thus, to the best of our knowledge, our results identify PyN L1CAM as the first known adhesion molecule required for the establishment and maintenance of neocortical ChC/PyN AIS innervation.

L1CAM is the founding member of the L1 family of CAMs, which also includes NrCAM, CHL1, and Neurofascin in vertebrates (Maness and Schachner, 2007). The L1CAM protein was initially described as a neuronal cell adhesion molecule involved in neurite/axon outgrowth (Appel et al., 1993; Lemmon et al., 1989; Rathjen and Schachner, 1984). Subsequent studies have also implicated L1CAM in processes like neuronal migration, axon guidance, synapse development/growth, and the arborization of presynaptic terminals (Castellani et al., 2000; Cohen et al., 1998; Enneking et al., 2013; Guan and Maness, 2010; Lindner et al., 1983). While these previous studies primarily focused on aspects of early development and/or the presynaptic function of L1CAM, our findings uncover a novel, previously unrecognized postsynaptic role for PyN L1CAM in regulating ChC/PyN AIS synaptic innervation/function in the neocortex. Importantly, this postsynaptic function of L1CAM is not compensated for by other L1 family members, as knockdown of L1CAM on its own is sufficient to cause the innervation phenotype. Moreover, depletion of neither NF186, NrCAM, nor CHL1 affected neocortical ChC/PyN AIS innervation. Our findings that L1CAM, but not NF186 or NrCAM, is required for innervation is intriguing, especially given that L1CAM is localized pan-axonally and not restricted to the AIS like NF186 and NrCAM. A likely explanation for this is that the presynaptic partner(s) of L1CAM on ChC axon terminals or in the surrounding environment is/are not recog-

nized by the other CAMs. At present, the identity of L1CAM's presynaptic binding partner(s) on ChC cartridges remains unknown. In this regard, a number of molecules have been reported to interact in *trans* with L1CAM but not with NF186 or NrCAM, including L1CAM itself, other Ig-domain containing proteins (e.g., ALCAM), extracellular matrix proteins (e.g., neurocan), and integrins (e.g., $\alpha_v\beta_3$, $\alpha_{IIb}\beta_3$) (Blaess et al., 1998; DeBernardo and Chang, 1996; Friedlander et al., 1994; Oleszewski et al., 1999). We can preclude the involvement of a *trans* L1CAM homophilic interaction, as silencing of L1CAM in ChCs did not affect ChC/PyN AIS innervation (data not shown). Exploring whether any other known L1CAM *trans*-interacting molecules are expressed on ChC cartridges or in the surrounding milieu and mediate the innervation of PyN AISs would be a worthwhile endeavor. Alternatively, we cannot exclude the possibility that a novel L1CAM ligand present on ChC cartridges or in the surrounding extracellular milieu mediates ChC/PyN AIS innervation. Future studies will be required to identify the presynaptic partner(s) of L1CAM necessary for proper neocortical ChC/PyN AIS innervation.

How does L1CAM govern selective PyN AIS innervation despite its pan-axonal localization? Our findings support a model in which the AIS AnkG- β IV-spectrin cytoskeletal complex anchors and clusters L1CAM at the AIS to promote high-affinity cell adhesion between ChC cartridges and PyN AISs, thereby facilitating axo-axonic synapse formation/stabilization. Indeed, we found that disruption of L1CAM-AnkG- β IV-spectrin interactions perturbs the subcellular distribution of surface L1CAM, reducing its enrichment at the AIS, and, most importantly, impairs PyN AIS innervation by ChCs. In contrast, the AnkG-free pool of L1CAM in the distal portion of the axon is likely more diffusible and not clustered (Boiko et al., 2007), rendering it incapable of mediating ChC/PyN synaptic connections. In line with our findings, previous work showed that cerebellar PC-expressed NF186 also requires association with AnkG to mediate AIS pinceau synapse formation (Ango et al., 2004), suggesting a conserved role for AnkG in the anchoring/clustering of L1 family members at the AIS. It is noteworthy, though, that while an AnkG-free form of NF186 at PC soma helps basket axon terminals navigate from the PC soma to the AIS, there is no indication that the AnkG-free form of L1CAM in the distal portion of the axon directs ChC cartridges from the distal axon of PyNs to their AIS. It is possible that direct innervation of neocortical PyN AISs by ChC cartridges occurs as a result of the unique, axonal arborization of ChCs. These cells undergo exuberant axonal overgrowth and branching followed by pruning (Inan and Anderson, 2014; Steinecke et al., 2017; Taniguchi et al., 2013); this is not the case for basket cells, which directionally orient their axon collaterals toward PC soma (Ango et al., 2004). In this regard, our data show that neocortical ChC axonal branching and complexity vastly increase at P11/P12, which, interestingly, coincides with an increase in PyN L1CAM expression at the AIS and the onset of ChC/PyN AIS innervation. Therefore, it is plausible that such morphometric changes bring ChC cartridges in close proximity to their target cells, allowing them to contact the AIS, bind AnkG-anchored L1CAM, and stabilize their connections. In contrast, the lesser-branched basket cell axons may require a molecular guide to help navigate to the AIS of

cerebellar PCs (Ango et al., 2004; Telley et al., 2016). We do not exclude the possibility, however, that another/other molecule(s) contribute(s) to ChC axon targeting to PyN AISs either by engaging in repulsive mechanisms at subcellular compartments other than the AIS or by functioning at the AIS itself where it may or may not act in *cis* with L1CAM. So far, though, we have found that, besides L1CAM, silencing of any of the other transmembrane adhesion molecules known to be localized at the AIS of PyNs did not disrupt neocortical ChC/PyN AIS innervation. Future work will be needed to fully elucidate the targeting mechanisms underlying neocortical PyN AIS innervation by ChCs.

Taken together, our findings unveil a critical role for PyN L1CAM, assembled by the AnkG- β IV-spectrin cytoskeletal complex at the AIS, in the axo-axonic innervation of neocortical PyNs by ChCs, thereby providing novel mechanistic insight into this unique form of subcellular innervation in the neocortex. Importantly, given our findings that L1CAM is required for proper neocortical ChC/PyN AIS innervation, this study also sheds light on the molecular mechanisms of connectivity defects underlying brain disorders. Interestingly, in this regard, mutations in L1CAM have been reported in patients with a broad spectrum of neurological abnormalities and mental retardation, which together are termed L1 syndrome (Jouet et al., 1994; Kenwrick et al., 1996; Weller and Gärtner, 2001). While major corpus callosum agenesis, retardation, adducted thumbs, shuffling gait, and hydrocephalus (CRASH) defects associated with various L1CAM mutations are likely due to perturbations in early developmental processes, such as neuronal migration and neurite outgrowth, several genetic alterations in the *L1CAM* gene are also associated with schizophrenia and epilepsy (Kurumaji et al., 2001; Poltorak et al., 1997; Weller and Gärtner, 2001), which commonly involve defects in neuronal connectivity. Thus, our findings that loss of L1CAM function causes postnatal defects in neocortical ChC/PyN AIS innervation and, ultimately, connectivity may provide a mechanistic explanation for some of the neurological aspects of L1 syndrome.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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● QUANTIFICATION AND STATISTICAL ANALYSIS

SUPPLEMENTAL INFORMATION

Supplemental Information can be found with this article online at <https://doi.org/10.1016/j.neuron.2019.02.009>.

A video abstract is available at <https://doi.org/10.1016/j.neuron.2019.02.009#mmc3>.

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AUTHOR CONTRIBUTIONS

Y.T., N.B.G., and L.V.A. conceived and/or designed the project. Y.T., N.B.G., M.W., and J.-R.Y. performed all the experiments, analyzed the data, and made the figures. L.V.A. wrote the paper with input from Y.T. and N.B.G. All authors edited and approved the manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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The following references appear in the Supplemental Information: Demyanenko et al., 2011, Greferath et al., 2002, Xu and Henkemeyer, 2009.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Chicken polyclonal anti-GFP	Aves Labs	Cat# GFP-1020; RRID: AB_10000240
Mouse monoclonal anti-AnkG (IgG2a)	NeuroMab	Cat# 75-146; RRID: AB_10673030
Mouse monoclonal anti-AnkG (IgG1)	NeuroMab	Cat# 75-187; RRID: AB_2057718
Rabbit polyclonal anti-Neurofascin-186	Gift from P.J. Brophy	N/A
Rabbit monoclonal anti-Neurofascin-186	Cell Signaling Technology	Cat# 15034S; RRID:AB_2773024
Rabbit polyclonal anti-RFP	Rockland	Cat# 600-401-379; RRID: AB_2209751
Goat polyclonal anti-tdTomato	Sicgen Antibodies	Cat# AB8181-200; RRID:AB_2722750
Rabbit polyclonal anti-βIV-Spectrin	Gift from M. Rasband	N/A
Rabbit polyclonal anti-NrCAM	Abcam	Cat# ab24344; RRID: AB_448024
Mouse monoclonal anti-Gephyrin (IgG1)	Synaptic Systems	Cat# 147011; RRID: AB_887717
Guinea pig polyclonal anti-VGAT	Synaptic Systems	Cat# 131004; RRID: AB_887873
Rabbit polyclonal anti-Nav1.6	Alomone Labs	Cat# asc-009; RRID: AB_2040202
Rat monoclonal anti-L1CAM	EMD Millipore	Cat# MAB5272; RRID: AB_2133200
Mouse monoclonal anti-L1CAM	Abcam	Cat# ab24704; RRID: AB_448241
Goat polyclonal anti-TAG-1	R&D Systems	Cat# AF4439; RRID: AB_2044647
Goat polyclonal anti-CHL1	R&D Systems	Cat# AF2147; RRID: AB_2079332
Mouse monoclonal anti-ADAM22	NeuroMab	Cat# 75-083; RRID:AB_10675128
Rabbit polyclonal anti-SCN1B	Abcam	Cat# ab107370; RRID: AB_10861904
Mouse monoclonal anti-γ-Tubulin	Sigma-Aldrich	Cat# T6557; RRID: AB_477584
Rabbit monoclonal anti-Vinculin	Cell Signaling Technology	Cat# 13901; RRID:AB_2728768
Mouse monoclonal anti-β-Actin	Sigma-Aldrich	Cat# A5441; RRID: AB_476744
Mouse monoclonal anti-Myc	EMD Millipore	Cat# 05-724; RRID: AB_11211891
Mouse monoclonal anti-HA.11	BioLegend	Cat# 901514; RRID: AB_2565336
Rabbit monoclonal anti-HA	Cell Signaling Technology	Cat# 3724; RRID: AB_1549585
Goat anti-chicken IgY Alexa Fluor 488	Thermo Fisher Scientific	Cat# A-11039; RRID: AB_142924
Goat anti-rabbit Alexa Fluor Plus 555	Thermo Fisher Scientific	Cat# A32732; RRID: AB_2633281
Goat anti-mouse IgG2a Alexa Fluor 647	Thermo Fisher Scientific	Cat# A-21241; RRID: AB_141698
Goat anti-mouse IgG1 Alexa Fluor 488	Thermo Fisher Scientific	Cat# A-21121; RRID: AB_141514
Donkey anti-goat Alexa Fluor 555	Thermo Fisher Scientific	Cat# A-21432; RRID: AB_141788
Donkey anti-mouse Alexa Fluor 647	Thermo Fisher Scientific	Cat# A-31571; RRID: AB_162542
Donkey anti-rabbit Alexa Fluor 594	Thermo Fisher Scientific	Cat# R37119; RRID: AB_2556547
Donkey anti-chicken IgY Alexa Fluor 488	Jackson ImmunoResearch	Cat# 703-545-155; RRID: AB_2340375
Anti-rabbit-HRP	Cell Signaling Technology	Cat# 7047; RRID: AB_2099233
Anti-mouse-HRP	Dako	Cat# P044701-2; RRID: AB_2617137
Anti-goat-HRP	Santa Cruz Biotechnology	Cat# sc-2350; RRID: AB_634811
Bacterial and Virus Strains		
One Shot Stbl3 Chemically Competent <i>E. Coli</i>	Thermo Fisher Scientific	Cat# C737303
Lentivirus: TRIPΔU3-EF1α-EGFP	Janas et al., 2006	N/A
Lentivirus: TRIPΔU3-EF1α-EGFP + H1-shL1CAM#1	This paper	N/A
Lentivirus: TRIPΔU3-EF1α-EGFP + H1-shL1CAM#2	This paper	N/A
Lentivirus: TRIPΔU3-EF1α-EGFP + H1-scramble control	This paper	N/A

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Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Chemicals, Peptides, and Recombinant Proteins		
Tamoxifen	Sigma-Aldrich	Cat# T5648; CAS: 10540-29-1
cOMplete, EDTA-free protease inhibitor cocktail	Sigma-Aldrich	Cat# 11873580001
Poly-D-Lysine	Sigma-Aldrich	Cat# A-003-E
Bovine Serum Albumin	Equitech-Bio, Inc.	Cat# BAH65
Tetrodotoxin	Tocris	Cat# 1078
6-cyano-7-nitroquinoxaline-2, 3-dione	Sigma-Aldrich	Cat# C127
DL-2-amino-5-phosphonopentanoic acid	Sigma-Aldrich	Cat# 074K38051
Critical Commercial Assays		
FuGENE HD	Roche	Cat# 4709691001
AMAXA Nucleofector Kit for mouse neurons	Lonza	Cat# VPG1001
Phusion High-Fidelity DNA Polymerase	New England Biolabs	Cat# M0530S
Takara DNA Ligation Kit Ver.2.1	Takara	Cat# 6022
Pierce™ BCA Protein Assay Kit	Thermo Fisher Scientific	Cat# 23225
Pierce™ ECL Western Blotting Substrate	Thermo Fisher Scientific	Cat# 32106
Experimental Models: Cell Lines		
HEK293T	ATCC	Cat# CRL-3216; RRID: CVCL_0063
Neuro-2a	ATCC	Cat# CCL-131; RRID: CVCL_0470
Experimental Models: Organisms/Strains		
Mouse: CD1	Charles River	Cat# CRL:22; RRID: IMSR_CRL:22
Mouse: CFW(Swiss Webster)	Charles River	Cat# CRL:24; RRID: IMSR_CRL:24
Mouse: Nkx2-1 ^{tm1.1(Cre/ERT2)Zjh/J}	Gift from Z.J. Huang	JAX: 014552; RRID: IMSR_JAX:014552
Mouse: B6;129S6-Gt(Rosa)26Sor ^{tm9(CAG-tdTomato)Hze/J}	Gift from Z.J. Huang	JAX: 007905; RRID: IMSR_JAX:007905
Oligonucleotides		
LacZ sgRNA: GTGCGAATACGCGGACGCGAT	This paper	N/A
Nf186 sgRNA, targeting exon 26: CCCCCGACGAGC AGTCCATT	This paper	N/A
L1CAM sgRNA, targeting exon 29 (C-terminal): TGGTGGACCTTGCTATTCT	This paper	N/A
ssODN sequence to integrate HA sequence into the genome just upstream of L1CAM's stop codon (HA sequence; underlined): ggcaagaaagagaaggaggcagcaggaggcaatgacagttc aggggctacctctcctatcaatcctgcagtagccctagaa <u>TACCCATACGATGTTCCAGATTACGCT</u> tagc aagggtccagccatgtgaggcagggccaaagctgggccaggg ccagagggtgcaggagagcccaggggccaagacacctggcc	This paper	N/A
For all shRNA sequences, see Table S2	N/A	N/A
Recombinant DNA		
Plasmid: pX330-U6-Chimeric_BB_Cbh_hSpCas9	Gift from F. Zhang	Addgene plasmid #42230 Cong et al., 2013
Plasmid: pX330-crNF186	This paper	N/A
Plasmid: pX330-crLacZ	This paper	N/A
Plasmid: pX330-crL1CAM	This paper	N/A
Plasmid: pSuper	OligoEngine	Cat# VEC-IND-0002
Plasmid: pSuper-shL1CAM#1	This paper	N/A
Plasmid: pSuper-shL1CAM#2	This paper	N/A
Plasmid: pSuper-shNrCAM	This paper	N/A
Plasmid: pSuper-shTAG-1	This paper	N/A

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Plasmid: pSuper-shCASPR2	This paper	N/A
Plasmid: pSuper-shEphA3	This paper	N/A
Plasmid: pSuper-shEphA4	This paper	N/A
Plasmid: pSuper-shEphB2	This paper	N/A
Plasmid: pSuper-shEfnB1	This paper	N/A
Plasmid: pSuper-shEfnB2	This paper	N/A
Plasmid: pSuper-shEfnB3	This paper	N/A
Plasmid: pSuper-shCtrl	This paper	N/A
Plasmid: pCAGGS (pCAG)	Watabe-Uchida et al., 2006	N/A
Plasmid: pCAG-MCS	This paper	N/A
Plasmid: pCAG-MCS-4	This paper	N/A
Plasmid: pCAG-miR	Tai et al., 2014	N/A
Plasmid: pCAG-miR.L1CAM	This paper	N/A
Plasmid: pCAG-miR.ADAM22	This paper	N/A
Plasmid: pCAG-miR.NF#1	This paper	N/A
Plasmid: pCAG-miR.NF#2	This paper	N/A
Plasmid: pCAG-miR.SCN1B	This paper	N/A
Plasmid: pCAG-miR.βIV#1	This paper	N/A
Plasmid: pCAG-miR.βIV#2	This paper	N/A
Plasmid: pCAG-miR.CHL1	This paper	N/A
Plasmid: pCAG-miR.Ctrl	This paper	N/A
Plasmid: pCAG-FLIP-miR	Tai et al., 2014	N/A
Plasmid: pCAG-FLIP-miR.L1CAM	This paper	N/A
Plasmid: pCAG-FLIP-miR.Ctrl	This paper	N/A
Plasmid: pCAG-Cre	Tai et al., 2014	N/A
Plasmid: pCAG-ER ^{T2} CreER ^{T2}	Gift from C. Cepko	Addgene plasmid #13777 Matsuda and Cepko, 2007
Plasmid: pCaMKII-miR	This paper	N/A
Plasmid: pCaMKII-miR.L1CAM	This paper	N/A
Plasmid: pCaMKII-miR.Ctrl	This paper	N/A
Plasmid: pcDNA3.1 mouse L1CAM	Gift from K. Kaibuchi	N/A
Plasmid: pcDNA3.1 human L1CAM	Gift from V. Lemmon	Addgene plasmid #12307 Hlavin and Lemmon, 1991
Plasmid: pcDNA3.1 human L1CAM Y1229H	Gift from P. Maness	N/A
Plasmid: pCAG human L1CAM	This paper	N/A
Plasmid: pCAG human L1CAM-Y1229H	This paper	N/A
Plasmid: HA-NF186	Gift from M. Rasband	N/A
Plasmid: pcDNA3.1 mouse CHL1	Gift from K. Kaibuchi	N/A
Plasmid: pcDNA3.1 rat SCN1B-Myc	This paper	N/A
Plasmid: pCMV6-XL4-NrCAM-FLAG	Gift from D. Montag	N/A
pCMV-Δ8.9	Janas et al., 2006	N/A
pVSVG	Janas et al., 2006	N/A
Plasmid: pTRIPΔU3-EF1α-EGFP	Gift from B. Verhasselt	N/A
Plasmid: pTRIPΔU3-EF1α-EGFP + H1- shL1CAM#1	This paper	N/A
Plasmid: pTRIPΔU3-EF1α-EGFP + H1-shL1CAM#2	This paper	N/A
Plasmid: pTRIPΔU3-EF1α-EGFP + H1-scramble control	This paper	N/A

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Software and Algorithms		
ImageJ	NIH	https://imagej.nih.gov/ij/ ; RRID: SCR_003070
LSM 5 Image Browser	Zeiss	https://www.embl.de/eamnet/html/body_image_browser.html ; RRID: SCR_014344
Zeiss Zen (Blue Edition) Imaging Software	Zeiss	https://www.zeiss.com/microscopy/en_us/products/microscope-software/zen.html#introduction ; RRID: SCR_013672
Axon pCLAMP 10 Electrophysiological Data Acquisition & Analysis Software	Molecular Devices	http://mdc.custhelp.com/app/answers/detail/a_id/18779/~/axon%E2%84%A2pclamp%E2%84%A2-10-electrophysiology-data-acquisition-%26-analysis-software-download ; RRID:SCR_011323
Mini Analysis Program	Synaptosoft	http://www.synaptosoft.com/MiniAnalysis/ ; RRID:SCR_002184
GraphPad Prism 7	GraphPad	http://www.graphpad.com/scientific-software/prism ; RRID: SCR_002798
Neurolucida	Microbrightfield Bioscience	https://www.mbfbioscience.com/neurolucida ; RRID: SCR_001775

CONTACT FOR REAGENT AND RESOURCE SHARING

Further information and requests for resources and reagents should be directed to and will be fulfilled by the Lead Contact, Linda Van Aelst (vanaelst@cshl.edu).

EXPERIMENTAL MODEL AND SUBJECT DETAILS

Animals

All experiments were carried out in accordance with the guidelines of the Animal Care and Use Committee of Cold Spring Harbor Laboratory (CSHL). *Nkx2.1-CreER* (JAX Stock 014552) and *Rosa26-loxpSTOPlox-TdTomato (Ai9)* (JAX Stock 007905) were provided by Z.J. Huang (CSHL). To genetically label a sparse population of neocortical chandelier cells (ChCs), we crossed *Nkx2.1-CreER^{+/−}* mice with *Rosa26-loxpSTOPlox-TdTomato (Ai9^{+/+})* reporter mice. To properly identify embryonic days 15.5 (E15.5) and E18.5 for *in utero* electroporations (IUEs) and tamoxifen (TMX) inductions, respectively, Swiss Webster females (Charles River) were housed with *Nkx2.1-CreER^{+/−};Ai9^{+/+}* males overnight and checked for a vaginal plug between 8:00–9:00 AM the next morning. Positive plug identification was designated E0.5. Timed-pregnant female CD1 mice (Charles River) were used for E13.5 and E15.5 double IUE experiments targeting ChCs and pyramidal neurons (PyNs), respectively. Mice were housed in groups of five per cage or singly if pregnant under standard vivarium conditions with a 12 h light/dark cycle. Animals were fed *ad libitum* and their health status were routinely monitored. Both male and female animals, except for SLENDR studies (see below), were used in all experiments.

HEK293T and Neuro-2a cell cultures

HEK293T (ATCC, RRID: CVCL_0063) and Neuro-2a cells (ATCC, RRID: CVCL_0470) were maintained in Dulbecco's Modified Eagle Medium (DMEM) (Thermo Fisher Scientific) containing 10% fetal bovine serum (FBS) (HyClone), 100 IU/mL penicillin (Thermo Fisher Scientific), and 100 µg/mL streptomycin (Thermo Fisher Scientific) in a humidified incubator at 37°C with 5% CO₂. The cell lines used in this study were not further authenticated and not found to be on the list of commonly misidentified cell lines (International Cell Line Authentication Committee).

Mouse cortical neuron culture

For the preparation of primary mouse cortical neuron cultures, cortices were dissected from E15.5 CD1 embryos of both sexes and digested in Hank's Balanced Salt Solution (HBSS) (Thermo Fisher Scientific) containing 0.25% trypsin (Sigma-Aldrich) and 0.1 mg/mL DNase (New England Biolabs) at 37°C for 15 min. Digested cortices were then resuspended in Minimal Essential Medium (MEM) (Thermo Fisher Scientific) containing 10% FBS, triturated with a glass Pasteur pipette, and filtered using a 70 µm cell strainer to obtain a single-cell suspension. The cell suspension was centrifuged at 1000 x g for 5 min at room temperature (RT) and the pellet was resuspended in MEM containing 10% FBS. Neurons were plated at a density of 1 × 10²/cm² on 12 mm glass coverslips (coated with 0.1 mg/mL poly-D-lysine (Sigma-Aldrich)) for immunofluorescence experiments or at a density of 2 × 10⁶ per well of a 6-well plate

(coated with 0.1 mg/mL poly-D-lysine) for western blot experiments. 2–4 h after plating, cell culture medium was replaced with Neurobasal medium (Thermo Fisher Scientific) supplemented with 2% B27 (Thermo Fisher Scientific), 100 IU/mL penicillin, 100 μ g/mL streptomycin, and 4 mM L-glutamine (Thermo Fisher Scientific). Neurons were maintained in a humidified incubator at 37°C with 5% CO₂. Half of the medium was replaced with fresh medium every 7 days.

METHOD DETAILS

Experimental Design

Animals were randomly allocated to experimental groups prior to experimental manipulation(s), and the experimenter was blinded to the condition until the experiment and quantifications/analyses were complete. For all IUE experimental conditions, analyses were performed on brains from a minimum of 3 electroporated animals. For all immunostaining experiments (*in vivo* and *in vitro*), quantifications/analyses were performed on cells from at least 3 individual animals. All western blotting experiments were performed in triplicate and depicted images are representative. No statistical methods were used to predetermine sample size in experiments. No data or animals were excluded from analysis.

Experimental constructs

The pCAG-MCS vector is a derivative of pCAGGS and was made by first blunting and religating the XhoI site of pCAGGS, thus abolishing the XhoI site, and then inserting a multiple cloning site (MCS) into the EcoRI site, abolishing the EcoRI site in the process. Human L1CAM and AnkG-binding mutant human L1CAM-Y1229H expression vectors (pCAG-hL1CAM and pCAG-hL1CAM-Y1229H, respectively) were generated by subcloning full-length L1CAM cDNA fragments from pcDNA3.1-hL1CAM (AddGene#12307) and pcDNA3.1-hL1CAM-Y1229H (gift from P. Maness), respectively, into the EcoRI and PmeI sites of pCAG-MCS-4, which is a derivative of pCAG-MCS with an EcoRI- and PmeI-containing MCS inserted into the PacI and NheI sites of pCAG-MCS. A full-length SCN1B cDNA fragment was obtained via RT-PCR of rat cortical mRNA (isolated from P14 rat cortices) and the cDNA was subcloned into the XhoI and EcoRI sites of pcDNA3.1/myc-His-B (Thermo Fisher Scientific). For RNA interference (RNAi) experiments, DNA fragments encoding control short hairpin RNA (shRNA) (Ctrl: 5'-GCTATACGGGATCGAAAGA-3') or shRNAs directed against mouse L1CAM (L1CAM#1: 5'-GCATCCACTTCAAACCCAA-3'; L1CAM#2 5'-AGCCTTACCAGAAGGGAAA-3'), mouse NrCAM (5'-GCAATGCCTCTAACAAATA-3'), mouse TAG-1 (5'-CCTGCTTTGCTGAGAACTT-3'), mouse CASPR2 (5'-GATTAGAGCCAGAGGGAAT-3'), mouse EphA3 (5'-GGACCTATGTTGATCCACATA-3'), mouse EphA4 (5'-GCAGCACCATCATCCATTG-3'), mouse EphB2 (5'-ACGAGAA CATGAACACTAT-3'), mouse ephrinB1 (5'-CACTGTGCTTGATCCCAAT-3'), mouse ephrinB2 (5'-GCAGACAGATGCACAATTA-3'), mouse ephrinB3 (5'-GCCTTCGGAGAGTCGCCAC-3'), mouse ErbB4 (5'-CCAGACTACCTGCAGGAATAC-3'), and mouse DOCK7 (5'-GGTACAGTACACATTTACA-3') were cloned into pSuper (Watabe-Uchida et al., 2006) via BglII and HindIII sites. The pCAG-miR vector backbone was made by inserting an EGFP-miR30 cassette (Stern et al., 2008) into the PacI and NheI sites of pCAG-MCS. DNA fragments encoding miR30-based shRNAs against Renilla luciferase (5'-AGGAATTATAATGCTTATCTA-3'), mouse L1CAM (5'-GAGAATCAACGGAATGTCTAA-3'), mouse CHL1 (5'-TACCAGGATAGAGGAAATTAA-3'), mouse ADAM22 (5'-ACATGG CAGATGTG ATCTATA-3'), mouse Neurofascin-186 (NF#1 5'-CACCAAGTCAATGCCATCTATA-3'; NF#2 5'-CACGATCTCGGTGA GAGTAAA-3'), mouse SCN1B (5'-GTCTACCGTCTCCTCTTCTTA –3'), and mouse β IV-spectrin (β IV#1 5'-CAGGAGAAATTCTCA GAGTTA-3'; β IV#2 5'-GACCACGATCGAGAAACTCAA-3') were cloned into the XhoI and EcoRI sites of pCAG-miR. The pCaMKII-miR vector backbone was constructed by replacing the CAG promoter of pCAG-miR with the CaMKII promoter via BamHI and PacI sites. pCaMKII-miR.L1CAM was constructed by inserting the miR30-based shRNA against mouse L1CAM into the XhoI and EcoRI sites of pCaMKII-miR. The pCAG-FLIP-miR vector backbone was described previously (Tai et al., 2014). pCAG-FLIP-miR.L1CAM was constructed by inserting the miR30-based shRNA against mouse L1CAM into the XhoI and EcoRI sites of pCAG-FLIP-miR. For lentiviral RNAi vectors (pTRIPΔU3-EF1 α -EGFP + H1-shL1CAM#1, #2, and scramble control), DNA fragments encoding shRNAs against mouse L1CAM (shL1CAM#1 or shL1CAM#2) or scramble control were first cloned into the pSuper(EcoRI) vector (Janas et al., 2006) via BglII and HindIII sites. The H1 promoter and shRNA sequences were then isolated from pSuper(EcoRI) via EcoRI digestion and subcloned into the EcoRI site of pTRIPΔU3-EF1 α -EGFP (Janas et al., 2006).

CRISPR construction

CRISPRs were designed at <http://zlab.bio/guide-design-resources> provided by the Zhang laboratory and then cloned into pX330 CRISPR/Cas9 vector (pX330-U6-Chimeric_BB-CBh-hSpCas9; a gift from Dr. Feng Zhang (Addgene plasmid #42230)) following Zhang's protocol (http://www.genome-engineering.org/crispr/?page_id=23). The target sequence of *Nf186* is CCCCCGACGAG CAGTCCATT, and the control (*LacZ*) target sequence is GTGCGAATACGGGACGCGAT.

In utero electroporation (IUE) and tamoxifen induction

To target/manipulate PyN gene expression and sparsely label ChCs in the same neocortical layer, ventricular zone (VZ)-directed IUE targeting neocortical progenitors was performed in *Nkx2.1-CreER;Ai9* embryos. Specifically, timed-pregnant Swiss Webster females that were bred with *Nkx2.1-CreER^{+/+};Ai9^{+/+}* males were anesthetized at 15.5 d of gestation, the uterine horns were exposed, and approximately 1 μ L of DNA solution (1 μ g/ μ L) was injected manually into the lateral ventricle of their embryos using a beveled and calibrated glass micropipette (Drummond Scientific). After injection, five square 50 ms pulses of 45 V with 950 ms intervals were

delivered across the uterus with two 5 mm electrode paddles (BTX, 45-0489) positioned on either side of the head (BTX, ECM830) (Figure 1A). After electroporation, the uterine horns were placed back in the abdominal cavity of the pregnant dam and the wound was surgically sutured. Tamoxifen (TMX; 3mg/30 g of body weight) was administered to the pregnant dam by oral gavage at 18.5 d of gestation to induce CreER activity and excision of the STOP cassette, resulting in tdTomato red fluorescent protein (RFP) expression in a sparse population of nascent neocortical ChCs in their offspring. Pups were sacrificed at postnatal days as indicated.

For experiments investigating whether L1CAM is required for the maintenance of neocortical ChC/PyN AIS innervation, CD1 embryos were subjected to both E13.5 and E15.5 IUEs targeting nascent ChCs and PyNs, respectively (Figure 5K). For E13.5 IUEs labeling nascent ChCs with tdTomato, 1 μ L of pCAG-tdTomato plasmid (1 μ g/ μ L) was injected manually into the lateral ventricle of the embryos and custom-made tweezer electrodes were placed at about 60° from the brain's horizontal plane so as to direct the current toward the ventral medial ganglionic eminence (vMGE) (Tai et al., 2014). Square electric pulses (35 V; 50 ms) were delivered five times at 950 ms intervals using an electroporator (BTX, ECM830). After electroporation, the uterine horns were placed back in the abdominal cavity of the pregnant dam and the wound was surgically sutured. Two days later, at E15.5, the same embryos were subjected to VZ-directed IUE with either an L1CAM-targeting or control Cre-dependent inducible RNAi expression vector (pCAG-FLIP-miR.L1CAM or pCAG-FLIP-miR.Ctrl, respectively) together with a vector expressing a TMX-inducible form of Cre recombinase (pCAG-ER^{T2}CreER^{T2}) (Matsuda and Cepko, 2007). TMX (3mg/30 g of body weight) was administered to the doubly electroporated animals via IP injection at postnatal day 28 (P28) to induce CreER activity and expression of miR.L1CAM or miR.Ctrl. The animals were sacrificed at P40 for analysis.

Transfection of HEK293T, Neuro-2a, and primary cultured cortical neurons

HEK293T cells were transfected using the calcium phosphate co-precipitation method. Neuro-2a cells were transfected using FuGENE HD (Promega) according to the manufacturer's instructions. Transfected cells were lysed 2 to 3 d after transfection and subjected to western blot analysis. Dissociated primary cortical neurons were transfected using the Amaxa Nucleofection system (Amaxa, Lonza) (Yang et al., 2012). In brief, 4×10^6 cells for each experimental condition were resuspended in 110 μ L electroporation buffer and transferred to an electroporation cuvette. Program O-005 was used to transfect the cells. After electroporation, cells were resuspended in MEM containing 10% FBS and plated at a density of 2×10^6 per well of a 6-well plate coated with 0.1 mg/mL poly-D-lysine. 2–4 h after plating, the medium was replaced with Neurobasal medium supplemented with 2% B27, 100 IU/mL penicillin, 100 μ g/mL streptomycin, and 4 mM L-glutamine. Cells were lysed 7 d (or 14 d in the case of AnkG and ADAM22) after transfection and subjected to western blot analysis. For immunofluorescence experiments, electroporated neurons were plated on 12 mm glass coverslips (coated with 0.1 mg/mL poly-D-lysine) at a density of $1 \times 10^2/\text{cm}^2$ and immunostained at days *in vitro* 10 (DIV10).

Lentivirus production and infection of primary neuronal cultures

Lentiviral supernatants were prepared and concentrated largely as described (Janas et al., 2006). Briefly, lentiviruses were produced by co-transfecting the transfer vector (pTRIP Δ U3-EF1 α -EGFP + H1-shL1CAM#1, #2, or scramble control), the HIV-1 packaging vector Δ 8.9 (pCMV- Δ 8.9), and the VSVG envelope glycoprotein vector (pVSVG) into HEK293T cells using the calcium phosphate co-precipitation method (Janas et al., 2006). Supernatants of culture media were collected 48 h after transfection and concentrated via ultracentrifugation at 100,000 $\times$ g for 2 h. The resulting pellet was resuspended in PBS, flash-frozen, and stored at -80°C . Primary mouse cortical neurons prepared at E15.5 (see above) were infected at DIV3, harvested at DIV10, and subsequently processed for western blot analysis.

Western blotting

Cells were washed with ice-cold PBS and lysed in RIPA lysis buffer containing 50 mM Tris-HCl, pH 7.4, 150 mM NaCl, 10 mM MgCl₂, 1% Triton X-100, 0.5% sodium deoxycholate, 0.1% SDS, 5% glycerol, 1 mM NaF, 1 mM Na₃VO₄, 1 mM PMSF, and 1x cComplete, EDTA-free protease inhibitor cocktail (Sigma-Aldrich). Protein concentration was determined using the Pierce BCA Protein Assay Kit (Thermo Fisher Scientific) according to the manufacturer's instructions. Cell lysates were resolved by SDS-PAGE and transferred to an Immobilon-P PVDF membrane (Millipore). Membranes were blocked in Tris-buffered saline with 0.5% Tween 20 (VWR) (TBST) containing 5% fat-free milk for 30 min at RT and then incubated in TBST with 5% bovine serum albumin (BSA) (Equitech-Bio, Inc.) and 0.05% NaN₃ containing primary antibodies overnight at 4°C. The following day, the membranes were incubated in blocking buffer containing horseradish peroxidase (HRP)-conjugated secondary antibodies for 1 h at RT. The following primary antibodies were used: anti-L1CAM (mouse mAb, 1:1000, Abcam ab24704), anti-TAG-1 (goat pAb, 1:1000, R&D Systems AF4439), anti-CHL1 (goat pAb, 1:1000, R&D Systems AF2147), anti- γ -tubulin (mouse mAb IgG1, 1:5000, Sigma-Aldrich T6557), anti- β -actin (mouse mAb IgG1, 1:9000, Sigma A5441), anti-vinculin (rabbit mAb, 1:2000, Cell Signaling Technology 13901T), anti-Myc (mouse mAb, 1:5000, EMD Millipore 05-724), anti-HA.11 (mouse mAb IgG1, 1:1000, BioLegend 901514), anti-NrCAM (rabbit pAb, 1:1000, Abcam ab24344), anti-ADAM22 (mouse mAb IgG2b, 1:200, NeuroMab 75-083), and anti-AnkG (mouse mAb IgG1, 1:200, NeuroMab 75-187). HRP-conjugated anti-rabbit (Cell Signaling Technology 7047), anti-mouse (Dako P044701-2), and anti-goat (Santa Cruz Biotechnology sc-2350) secondary antibodies were used at 1:5,000. Pierce ECL Western Blotting Substrate (Thermo Fisher Scientific) was used for detection of HRP activity.

Immunohistochemistry and immunocytochemistry

For immunostaining of tissue sections, animals were deeply anesthetized with isoflurane and perfused transcardially with PBS and 4% paraformaldehyde (PFA) in 0.1 M phosphate buffer. Brains were post-fixed in 4% PFA in 0.1 M phosphate buffer overnight at 4°C and then cryoprotected with 30% sucrose in PBS. Of note, for neurofascin-186, NrCAM, and Nav1.6 immunostaining, animals were perfused and post-fixed in 1% PFA in 0.1 M phosphate buffer prior to cryoprotection. 50 μ m thick coronal sections were subsequently generated using a Vibratome (Leica VT1000S). For gephyrin, TAG-1, and SCN1B immunostaining, antigen retrieval was performed prior to the application of primary antibodies. Namely, brain sections were incubated at 95°C in antigen retrieval buffer (25 mM Tris-HCl, pH 8.5; 1 mM EDTA, 0.05% SDS) for 15 min. For immunostaining of endogenous L1CAM in brain slices, fresh brains were flash frozen in liquid nitrogen, cryosectioned at 12 μ m thickness using a cryostat (Leica CM1850), and mounted onto Superfrost Plus microscope slides (Fisherbrand). Mounted brain sections were then thawed at RT and briefly fixed in 2% PFA in PBS for 90 s. Following the aforementioned methods of tissue fixation and sectioning described for each immunohistochemistry (IHC) experiment, brain slices were blocked and permeabilized with 10% normal goat serum (NGS) or normal donkey serum (NDS) and 0.3% Triton X-100 in PBS at RT for 1 h and then incubated with primary antibodies diluted in 3% NGS or NDS and 0.3% Triton X-100 in PBS overnight at 4°C. Fluorescently-conjugated secondary antibodies diluted in 3% NGS or NDS and 0.3% Triton X-100 in PBS were applied for 1 h at RT the following day. For live cell surface labeling of L1CAM on DIV10 cortical neurons, cortical neuron cultures were washed twice with PBS and live-labeled with an anti-L1CAM antibody (1:150) diluted in PBS for 15 min at 37°C. The cultures were then fixed with 4% PFA in 0.1 M phosphate buffer for 15 min at 4°C, permeabilized with 0.1% Triton X-100 in PBS for 5 min, blocked with 10% NGS in PBS for 1 h at RT, and subsequently incubated with anti-GFP (1:1000) and anti-AnkG (1:1000) antibodies diluted in 3% NGS in PBS overnight at 4°C. The next day the cultures were incubated with fluorescently-conjugated secondary antibodies (1:2000) diluted in 3% NGS in PBS for 1 h at RT. The following primary antibodies were used: anti-GFP (chicken pAb, 1:1000, Aves Labs GFP 1020), anti-AnkG (mouse mAb IgG2a, 1:500 (IHC) or 1:1000 (immunocytochemistry), NeuroMab 75-146), anti-Neurofascin-186 (rabbit pAb, 1:500, gift from P. Brophy or rabbit mAb, 1:500, Cell Signaling Technology 15034), anti-RFP (rabbit pAb, 1:1000, Rockland 600-401-379), anti-tdTomato (goat pAb, 1:100, Sicgen Antibodies AB8181-200), anti- β IV-spectrin (rabbit pAb, 1:500, gift from M. Rasband), anti-NrCAM (rabbit pAb, 1:500, Abcam ab24344), anti-Gephyrin (mouse mAb IgG1, 1:500, Synaptic Systems 147011), anti-VGAT (guinea pig pAb, 1:500, Synaptic Systems 131004), anti-Nav1.6 (rabbit pAb, 1:100, Alomone Labs asc-009), anti-TAG-1 (goat pAb, 1:200, R&D Systems AF4439), anti-SCN1B (rabbit pAb, 1:100, Abcam ab107370), anti-HA (rabbit mAb, 1:500, Cell Signaling Technology 3724), and anti-L1CAM (rat mAb, 1:150, EMD Millipore MAB5272). The following secondary antibodies were used: Alexa Fluor (AF) 488 goat anti-chicken IgY (1:1000, Thermo Fisher Scientific A-11039), AF Plus 555 goat anti-rabbit (1:1000, Thermo Fisher Scientific A32732), AF 647 goat anti-mouse IgG2a (1:1000, Thermo Fisher Scientific A-21241), AF 488 goat anti-mouse IgG1 (1:1000, Thermo Fisher Scientific A-21121), AF 555 donkey anti-goat (1:1000, Thermo Fisher Scientific A-21432), AF 647 donkey anti-mouse (1:1000, Thermo Fisher Scientific A-31571), AF 594 donkey anti-rabbit (1:1000, Thermo Fisher Scientific R37119), and AF 488 donkey anti-chicken (1:1000, Jackson ImmunoResearch 703-545-155).

In vivo single-cell labeling of endogenous L1CAM by CRISPR-Cas9-mediated homology-directed repair (SLENDR)

As a complementary approach to *in vivo* endogenous L1CAM immunostaining, we took advantage of SLENDR, a CRISPR-Cas9-mediated homology-directed repair genome editing strategy (Mikuni et al., 2016), to insert an HA epitope tag into the genomic locus of *L1CAM* and subsequently visualize the protein's subcellular localization via HA immunostaining at single-cell resolution *in vivo*. Specifically, a single-guide RNA (sgRNA) (5'-TGGTGGACCTTGCTATTCT-3') targeting the vicinity of L1CAM's stop codon was designed and cloned into the pX330 CRISPR/Cas9 vector following Zhang's protocol (http://www.genome-engineering.org/crispr/?page_id=23) (pX330-crL1CAM). In addition, a corresponding single-stranded oligodeoxynucleotide (ssODN-L1CAM) (5'-ggcaagaaagagaaggaggcaggaggcaatgacagttcaggggctacgtctctcatcaatctcagtcagtcagccctagaaTACCCATACGATTGCCAGATTACGCTtagcaagggtccagccatgtgaggcagggcccaagctgggccaggccagaggtgcaggagagccaggggccaagacacctggcc-3'; HA sequence underlined) was designed to mediate the integration of the HA epitope tag sequence into the genome immediately upstream of L1CAM's stop codon. A cocktail of pX330-crL1CAM (1 μ g/ μ l), ssODN-L1CAM (20 μ M), and pCAG-YFP (1 μ g/ μ l) was then used in E13.5 IUEs targeting PyN progenitors, as described above. Given that *L1CAM* is located on the X chromosome, only electroporated male pups were analyzed to avoid any complications arising from differences in the number of alleles edited. Male pups were sacrificed at P10 and P13 and brain slices from such animals were immunostained (as described above) with anti-HA and anti-AnkG antibodies.

Preparation of acute brain slices and electrophysiology

CD1 mice (P27–30) were anaesthetized with isoflurane, decapitated, and their brains quickly removed and chilled in ice-cold dissection buffer (110 mM choline chloride, 25 mM NaHCO₃, 1.25 mM NaH₂PO₄, 2.5 mM KCl, 0.5 mM CaCl₂, 7 mM MgCl₂, 25 mM glucose, 11.6 mM ascorbic acid, and 3.1 mM pyruvic acid; gassed with 95% O₂ and 5% CO₂). Coronal slices (300 μ m) containing the somatosensory cortex were cut in dissection buffer using a Vibratome (Leica VT1000S) and subsequently transferred to a storage chamber containing artificial cerebrospinal fluid (ACSF; 118 mM NaCl, 2.5 mM KCl, 26.2 mM NaHCO₃, 1 mM NaH₂PO₄, 20 mM glucose, 2 mM MgCl₂, and 2 mM CaCl₂ at 34°C, pH 7.4; gassed with 95% O₂ and 5% CO₂). After at least 40 min recovery time, slices were transferred to RT and constantly perfused with ACSF.

Whole-cell patch-clamp recordings were performed on electroporated GFP+ PyNs in layer II/III in the somatosensory cortex. Recordings were obtained with Multiclamp 700B amplifiers (Molecular Devices) under visual guidance using a Zeiss Axioskop

microscope equipped with both transmitted light illumination and epifluorescence illumination. Miniature inhibitory postsynaptic currents (mIPSCs) were recorded in voltage-clamp mode with borosilicate pipettes (3.5–5 M Ω). The internal solution (ACSF) contained the following: 115 mM cesium methanesulphonate, 20 mM CsCl, 10 mM HEPES, 2.5 mM MgCl₂, 4 mM Na₂-ATP, 0.4 mM Na₃GTP, 10 mM Na-phosphocreatine, and 0.6 mM EGTA (pH 7.2). mIPSCs were recorded at a holding potential of 0 mV with 6-cyano-7-nitroquinoxaline-2, 3-dione (CNQX, 20 μ M), DL-2-amino-5-phosphonopentanoic acid (DL-AP5, 100 μ M), and tetrodotoxin (TTX; 1 μ M) added to the ACSF. Electrophysiological data were acquired using Axon pCLAMP 10 software (Molecular Devices) and analyzed using Mini Analysis Program (Synaptosoft).

Confocal image acquisition and analysis of ChC/PyN percent innervation and inhibitory synapse number

For analysis of ChC/PyN AIS percent innervation, images of coronal brain slices (50 μ m thickness) were acquired using an LSM 800 confocal laser-scanning microscope (Zeiss) with a 63x oil-immersion objective and sequential acquisition settings applied at a resolution of 1024x1024 pixels. 200 μ m x 200 μ m images of single RFP+ ChCs and neighboring GFP+ electroporated PyNs in layer II of the somatosensory cortex were collected from E18.5 TMX-induced *Nkx2.1-CreER^{+/+};Ai9^{+/-}* mice (at postnatal ages as indicated) using a z series of 40–50 images with a depth interval of 1 μ m. ChC/PyN AIS percent innervation was calculated by dividing the number of GFP+ PyNs innervated by a single RFP+ ChC by the total number of GFP+ PyNs present in the entire 200 μ m x 200 μ m image z stack. Average ChC cartridge number was determined for PyN L1CAM and control knockdown conditions by quantifying the number of RFP+ cartridges from single RFP+ ChCs in 200 μ m x 200 μ m confocal images acquired from *Nkx2.1-CreER^{+/+};Ai9^{+/-}* coronal brain slices (50 μ m thickness) as described above. Importantly, TMX administration at E18.5 was essential for standardizing our analyses of ChC/PyN AIS percent innervation to single ChCs, since such an induction protocol enables the sparse labeling of spatially isolated neocortical ChCs in *Nkx2.1-CreER^{+/+};Ai9^{+/-}* animals. Of note, TMX induction even 24 hours earlier (at E17.5) causes a significantly larger population of overlapping and non-homogeneously distributed ChCs to be labeled in the neocortex (Figure S5A), which causes a marked increase in variability of ChC/PyN AIS percent innervation values across different images analyzed (Figure S5B). Notably, variations in the number of electroporated GFP+ PyNs present in each field of view (FOV) did not impact ChC/PyN AIS percent innervation (Figure S5C). Moreover, “convex hull” analysis (Inan et al., 2013) of ChC/PyN AIS percent innervation was performed on serial reconstructions of entire individual RFP+ ChCs and all neighboring GFP+ PyNs within the domain of the ChC axonal arbor. Such reconstructions were generated from serial 320 μ m x 320 μ m z stack images acquired from four to six 50 μ m thick brain slices using an LSM 800 confocal laser-scanning microscope (Zeiss) equipped with an 40x oil-immersion objective. Importantly, this analysis recapitulated the PyN L1CAM knockdown innervation phenotype relative to the control condition obtained in the 200 μ m x 200 μ m FOV (Figure S5D). Given this, all analyses of ChC/PyN AIS percent innervation were performed on 200 μ m x 200 μ m z stacks of 40–50 images.

Representative maximum projection images of ChC/PyN AIS innervation with gephyrin or VGAT staining were generated using a z series of 4–8 images with a depth interval of 0.37 μ m. The average number of gephyrin or VGAT puncta per PyN AIS was determined manually. The percentage of RFP+ ChC boutons overlapping with AIS gephyrin puncta was calculated by dividing the number of RFP+ ChC boutons overlapping with gephyrin puncta by the total number of RFP+ ChC boutons per AIS. The average number of gephyrin or VGAT puncta per μ m on PyN dendrites was quantified by manually counting the number of gephyrin or VGAT puncta within 30 μ m of the cell body on the apical dendrite and dividing this sum by the length of the dendrite analyzed. To quantify the average number of VGAT puncta per μ m on PyN cell bodies, single z-plane images were acquired using an LSM 800 confocal laser-scanning microscope (Zeiss) with a 63x oil-immersion objective and acquisition settings applied at a resolution of 1024 x 1024 pixels. The number of VGAT puncta overlapping with GFP+ PyN cell bodies was manually counted and the circumference of the cell bodies was measured using Zeiss Zen (Blue Edition) imaging software. The average number of VGAT puncta per μ m on PyN cell bodies was then calculated by dividing the number of PyN cell body VGAT puncta by the circumference of the cell body. For quantification of the average fluorescence intensity (a.u.) of PyN perisomatic gephyrin signal, the “integrated fluorescence intensity” (the sum of the fluorescence intensity values of all pixels in a region of interest (ROI)) was calculated in a perisomatic ring (an intracellular ring ROI spanning 2 μ m from the cell membrane) in a single z-plane image using ImageJ software. The “mean fluorescence background” was also measured by selecting an area(s) outside of the cell and next to each ROI that has no fluorescence. The average corrected fluorescence intensity of perisomatic gephyrin was then calculated as: ((Integrated fluorescence intensity of perisomatic gephyrin) – (Area of ROI x Mean fluorescence intensity of background readings))/Area of ROI.

Confocal image acquisition and analysis of endogenous and surface L1CAM and other AIS proteins

For quantification of endogenous L1CAM or L1CAM-HA expression levels *in vivo*, images of brain slices immunostained for L1CAM or HA (see above) were acquired using an LSM 800 confocal laser-scanning microscope (Zeiss) with a 63x oil-immersion objective and sequential acquisition settings applied at a resolution of 1024x1024 pixels. Representative maximum projection images of PyN AISs were generated using a z series of 4–8 images with a depth interval of 0.37 μ m. Relative fluorescence intensity of AIS-localized L1CAM or L1CAM-HA was calculated using ImageJ software. Namely, the relative fluorescence intensity was calculated by first measuring the “integrated fluorescence intensity” of L1CAM on the entire AIS in spatially isolated PyNs in addition to the “mean fluorescence background” (as described above). The relative corrected fluorescence intensity was then calculated as: ((Integrated

fluorescence intensity of L1CAM) – (Area of the AIS x Mean fluorescence intensity of background readings))/Area of AIS normalized to P10. The relative fluorescence intensities of AIS-enriched proteins under PyN L1CAM knockdown conditions were calculated as described above and normalized to control conditions.

For surface L1CAM staining in cultured primary cortical neurons, images of randomly chosen GFP+ transfected cells were acquired using an LSM 800 confocal laser-scanning microscope (Zeiss) with a 40x oil-immersion objective. Sequential acquisition settings were applied at a resolution of 1024 × 1024 pixels. For representative images of GFP+ cultured cortical neurons, maximum projections of 4-6 sequential z stack images at a 0.6 μm depth interval were generated. To calculate average surface L1CAM fluorescence intensity (a.u.), ImageJ software was used to measure the “integrated fluorescence intensity” of surface L1CAM on the AIS or distal axon (within a region of the axon spanning approximately 40 μm from the AIS) of an isolated GFP+ neuron as well as the “mean fluorescence background” (as described above). The average corrected fluorescence intensity of surface L1CAM was then calculated as: ((Integrated fluorescence intensity of surface L1CAM) – (Area of ROI x Mean fluorescence intensity of background readings))/Area of ROI. The average corrected fluorescence intensity of GFP for each ROI was also calculated in the same manner and the average corrected fluorescence intensity of surface L1CAM was then normalized to the average corrected fluorescence intensity of GFP for each ROI by dividing the average corrected fluorescence intensity of surface L1CAM by the average corrected fluorescence intensity of GFP. In addition, the ratio of GFP-normalized surface L1CAM fluorescence intensity at the AIS to GFP-normalized surface L1CAM fluorescence intensity at the distal axon was calculated per cell and averaged for each condition. AIS length and area were measured using the appropriate measurement tools in Zeiss Zen (Blue Edition) imaging software and ImageJ, respectively.

ChC axonal 3D reconstruction and morphometric analyses

For immunostaining of tissue sections for ChC axonal 3D reconstructions and morphometric analyses, brains from E18.5 TMX-induced *Nkx2.1-CreER^{+/+};Ai9^{+/+}* animals were collected at the postnatal ages indicated and processed/sectioned and immunostained as described above. Immunostained brain sections were subsequently imaged using an LSM 800 confocal laser-scanning microscope (Zeiss) with a 63x oil-immersion objective and sequential acquisition settings applied at a resolution of 1024x1024 pixels. Of note, for all 3D reconstructions, 200 μm x 200 μm images of single RFP+ ChCs, which were spatially isolated from other RFP+ ChCs and processes, were collected in the somatosensory cortex using a z series of 40-50 images with a depth interval of 1 μm. ChC axonal arbors were reconstructed using the User-Guided tracing mode of Neurolucida 360 (Microbrightfield Bioscience). Morphometric analyses of ChC axonal complexity, length, and the number of branch points as well as ChC axonal Sholl analysis were carried out using Neurolucida Explorer (Microbrightfield Bioscience). ChC axonal complexity index was defined as: ($\sum$ terminal orders + number of terminals) x (total axonal length) where the number of “terminal orders” for each terminal point is calculated as the number of branches that appear in proceeding backward from the defined terminal to the ChC soma. ChC “terminals” (or axonal endings) were defined as the smallest RFP+ ChC axonal processes clearly identifiable from the last branching point in high resolution confocal images.

QUANTIFICATION AND STATISTICAL ANALYSIS

For quantifications/analyses of ChC/PyN percent innervation, 5-29 ChCs and 4-630 PyNs per ChC from at least 3 animals were analyzed. For immunostaining and SLENDR-based fluorescence intensity experiments, 14-155 AISs from 3 animals (*in vivo*) or 35-43 neurons from 3 independent cultures (from at least 3 individual animals) (*in vitro*) were analyzed. Of note, for *in vivo* and *in vitro* immunostaining experiments, all depicted images are representative. For quantifications/analyses of PyN AIS length and area, 14-40 AISs from 3 animals were analyzed and for average cartridge number per ChC, 10 ChCs from 3 animals were analyzed. Moreover, for gephyrin and VGAT immunostaining analysis, 8-49 AISs, 30-34 cell bodies, or 24-51 dendrites from at least 3 animals were analyzed. 16 GFP+ PyNs from 4 animals were analyzed for electrophysiological measurements of mIPSC amplitude and frequency. For ChC soma and axon 3D reconstructions and morphometric analyses, maximum projection renderings are representative and 4-7 ChCs from 3 animals were analyzed. Lastly, all western blot experiments were performed in triplicate and depicted images are representative. Exact sample sizes for all experiments are indicated in their respective figure legends.

All data are presented as mean ± SEM. Differences between experimental conditions were analyzed with GraphPad Prism 7 software (GraphPad). Comparisons of two groups were made using unpaired two-tailed Student's t tests. For comparisons of more than two groups, one-way analysis of variance (ANOVA) with post hoc Tukey-Kramer multiple comparison tests were performed. Statistical significance was defined as $p < 0.05$, $p < 0.01$, or $p < 0.001$ (indicated as *, **, or ***, respectively). p values ≥ 0.05 were considered not significant.